INCH-POUND

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DETAIL SPECIFICATION

FILTER AND FILTER ELEMENTS, FLUID PRESSURE, HYDRAULIC LINE, 15 MICRON AND 5 MICRON, TYPE II, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hydraulic filter assemblies and filter elements that retain all particles larger than 15 microns and 5 microns and are for use as specified in 6.1 and the applicable specification sheet.

1.2 Classification.

1.2.1 Type. Hydraulic filter assemblies are of the following type:

Type II	-	-65 ºF to +275 ºF fluid temperature range as defined in SAE-AS5440.

1.2.2 Classes. Hydraulic filter assemblies are of the following classes:

Class 1500	-	Filter assemblies and filter elements furnished in accordance with this specification are for operating pressures up to and including 1,500 pounds per square inch (psi), as defined in SAE-AS5440.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop B120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to lke-navairdoccom.fct@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Class 3000 - Filter assemblies and filter elements furnished in accordance with this specification are for operating pressures up to and including 3,000 psi, as defined in SAE-AS5440.

Class 4000 - Filter assemblies and filter elements furnished in accordance with this specification are for operating pressures up to and including 4,000 psi, as defined in SAE-AS5440.

Class 5000 - Filter assemblies and filter elements furnished in accordance with this specification are for operating pressures up to and including 5,000 psi, as defined in SAE-AS5440.

1.2.3 Styles. Hydraulic filter assemblies are of the following styles:

Style A - Bypass housing incorporates provision for bypassing fluid at 100 ±10 psi or as required in accordance with the applicable specification sheet.

Style B - Non-bypass housing with no provisions for bypassing fluid.

1.2.4 Sizes and micron ratings. Micron ratings and sizes are as specified in the applicable specification sheets (see 6.2). The hydraulic filter element is either cleanable or non-cleanable, as specified on the applicable specification sheet.

1.2.5 Part number. Part numbers are as specified in the applicable specification sheets.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680 - Degreasing Solvent

MIL-PRF-5606 - Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance (Inactive for new design)

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MIL-PRF-32295	-	Cleaner, Non-Aqueous, Low-Voc, Hap-Free
MIL-PRF-81836	-	Filter and Disposal Element, Fluid Pressure, Hydraulic 3 Micron Absolute, General Specification for
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-87257	-	Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
AN929	-	Cap Insert Assembly, Pressure Seal, Flared Tube Fitting
MS21344	-	Fittings, Installation of Flared Tube, Straight Threaded Connectors, Design Standard For

(See supplement 1 for list of specification sheets.)

(Copies of these documents are available online at https://assist.dla.mil.)

2.2.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAIR DRAWINGS

63A109-D-78	-	Fixture Test Assembly
63A109-C-91	-	Plumbing Schematic, Operational Hydraulic Filter Cleaner

(Copies of these drawings can be obtained from JEDMICS with an account.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SAE INTERNATIONAL

SAE-AIR787	-	Filter Element Cleaning Methods

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SAE-ARP785	-	Procedure for the Determination of Particulate Contamination in Hydraulic Fluids by the Control Filter Gravimetric Procedure (DoD adopted)
SAE-AS598	-	Aerospace Microscopic Sizing and Counting of Particulate Contamination for Fluid Power Systems (DoD adopted)
SAE-AS4059	-	Aerospace Cleanliness Classification for Hydraulic Fluids (DoD adopted)
SAE-AS5174	-	Fitting, Union and Reducer, Flared (DoD adopted)
SAE-AS5440	-	Hydraulic Systems, Aircraft, Design and Installation Requirements for (DoD adopted)
SAE-AS8775	-	Hydraulic Systems Components, Aircraft and Missiles, General Specification for (DoD adopted)

(Copies of these documents are available for purchase online at www.sae.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection By Attributes (DoD adopted)

(Copies of this document are available for purchase online at www.asq.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. The filter assemblies and filter elements furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.4 and 6.3). In addition, the retention of the qualification for the filter assembly and filter element on the applicable qualified products list shall be dependent on periodic verification of continued compliance with the requirements of this specification (see 4.3.4).

3.2 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.3 General specification. The requirements of SAE-AS8775 apply as requirements of this specification with the exception and additions specified herein. When the two specifications conflict, this specification shall govern.

3.4 Materials.

3.4.1 Compatibility. The filter assemblies and filter elements shall be constructed of materials that shall not adversely affect or be affected by hydraulic fluid conforming to MIL-PRF-5606, MIL-PRF-83282, MIL-PRF-87257, and other test fluids included herein.

3.4.2 Cleanable filter elements. Cleanable filter elements shall be constructed of materials that shall not be adversely affected by the cleaning procedures specified in 4.5.2.10.2.3. Cleanable filters elements shall not be used in aerospace-based hydraulic applications.

3.4.3 Process controls limitations. If the filter element is of the type wherein resinous or other material is used for elimination of imperfections, not more than 5 percent of the filtering area shall be covered by the applied material. Filter element media having an initial bubble point of less than 3 inches of water before repair is cause for rejection.

3.4.4 Filter element repair procedures (for cleanable elements only). Filter element repair procedures shall specify, but not be limited to, the resin for joining or patching, procedures for resin adhesion, application techniques, curing cycle, special tools and test equipment required.

3.4.4 Aging. Filter element materials that are subjected to aging shall be fully defined.

3.4.5 Fungi nutrients. Materials that are not nutrients for fungi shall be used wherever possible. Where fungus-nutrient materials are used, they shall be treated with a fungicidal agent acceptable to the qualifying activity.

3.4.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.5 Design and construction.

3.5.1 Filter assemblies and filter elements. Filter assemblies and filter elements, for hydraulic systems conforming to SAE-AS5440 shall be designed and constructed as specified herein and in the applicable specification sheet. Filter assemblies shall be designed and constructed so that the elements may be removed for service and inspection without disconnecting fittings or disturbing mountings. Filter assemblies shall be designed so that fluid entering the filter housing cannot impinge directly upon element filter medium or upon the relief valve poppet. The filter assemblies shall be capable of withstanding 2,000 hours of operation within the temperature range defined in 1.2.1. The manufacturer shall verify compliance with this requirement by analysis. Differential pressure indicators and pressure relief valves (except for the valve seat) shall be of modular design and capable of being readily removed from the filter assembly for replacement.

3.5.2 Flow type. Filter assemblies and filter elements shall be of the full-flow type. The flow through the filter elements shall be from "outside-in."

3.5.3 Construction. The design of the filter assemblies shall be such that the filter element cannot be improperly installed.

3.5.4 Filter housing design. Filter housings shall be designed to withstand all the structural loads imposed by the functional test requirements of this specification. In addition, the filter housing and mountings shall be of such strength and rigidity as to withstand the wrench loads required for making tube connections in accordance with MS21344 and the replacing of filter elements.

3.5.5 Interchangeability. It shall not be possible to interchange filter bowls and heads between the various sizes of filter assemblies, except that -4, -6, and -8 sizes shall use the same bowl.

3.5.6 Protective devices.

3.5.6.1 Filter assembly relief valves. Style A filter assemblies shall incorporate a relief valve to permit bypassing of the hydraulic fluid in the event of excessive restriction to flow through the filter element. The relief valve cracking pressure shall be 100 ±10 psi or as required in accordance with the applicable specification sheet. Reseat pressure shall be 65 psi minimum or as required in accordance with the applicable specification sheet (see 4.5.1.4.1). These pressures apply over the full temperature range of -65 ºF to +275 ºF. No relief valve shall be incorporated in style B filter assemblies.

3.5.6.1.1 Relief valve design. The relief valve design shall be such that no malfunction can occur due to flow surges up to 150 percent of the rated flow of the filter. At 150 percent of rated flow through the relief valve and at 100 ºF ±10 ºF fluid temperature, the differential pressure across the relief valve shall not exceed 240 psi.

3.5.6.1.2 Relief valve type. The relief valve design shall be of the non-adjustable type.

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3.5.6.1.3 Filter assembly location. The filter assembly shall have the bowl installed below horizontal with the relief valve located above to prevent inborne contaminants from accumulating or settling in the relief valve.

3.5.6.2 Filter element relief valves. Filter element pressure relief valve, when required by the applicable specification sheet, shall meet the requirements of the applicable specification sheet and the requirements and tests specified herein (see 4.5.2.12).

3.5.6.3 Differential pressure indicator. Filter assemblies shall incorporate a differential pressure indicator (DPI) that will give visual warning by raising a mechanical red indicator or lighting a device when the differential pressure across the filter element exceeds 70 pounds per square inch differential (psid) ±10 percent for mechanical DPI or 90 psid ±2 percent for digital DPI. Electronic DPIs shall have an additional yellow indication at 70 psid ±2 percent. If an electronic DPI incorporates a temperature sensing device into the design, the device shall indicate when hydraulic oil temperature has exceeded 260 °F ±10 ºF. A temperature sensing device is meant to be used for maintenance purposes. Inadvertent actuating shall not occur under a temporary flow surge or a 20g shock load. If electronic, the DPI shall be programmed to reject spikes of pressure (less than 100 milliseconds) of duration, due to shock or water hammer effect. Only stable readings representative of the filter differential pressure will be monitored. Monitoring of the pressure and temperature are to be programmed into the sensor. External contamination shall not adversely affect the normal operation of the differential pressure indicator and positive protection against external contaminants shall be provided. If a warning light is used, the indicator shall provide means to check for normal operation on the ground, such as a “push to test” button. This warning light shall be capable of operation for up to one year before failure. Once actuated, the mechanical red indicator or warning light shall remain extended or warning light shall remain illuminated until reset manually. When the indicator is in the reset or normal position, the mechanical red indicator or light shall be hidden from view or not illuminated. The indicator shall not actuate or illuminate below 100 °F ±15 °F fluid temperature. If a mechanical indicator is used, it shall be red in color. If a digital warning light is used, the following light signals shall be incorporated into the design:

Differential Pressure greater than or equal to 90 psid ±2%: RED (Remain Illuminated)
Differential Pressure greater than or equal to 70 psid ±2%: YELLOW (Remain Illuminated)
Differential Pressure below 68 psid: GREEN or Normal (Push to Test for Normal)

If a temperature sensing device is incorporated into the design:

Temperature exceeds 260 ºF ±10 ºF: RED (Remain Illuminated)

3.5.6.4 Automatic shutoff. Styles A and B filter housings shall be provided with an automatic shutoff device to prevent drainage from the flow between the inlet and outlet ports of the filter head when the filter bowl and element are removed, and to prevent air inclusion in excess of 2 cubic centimeters (cc) upon reassembly. The device for style B filter housings, shall allow rated flow through the filter housing at a differential pressure not to exceed 200 psi in the event that the filter element is not installed in a filter assembly. The device shall pass the leakage, cycling, and air inclusion tests specified under 4.5.1.4.3.

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3.6 Performance.

3.6.1 Filter assemblies. Filter assemblies including filter housings, and filter elements shall perform satisfactorily when subjected to the tests specified herein.

3.6.2 Impulse. Filter assemblies shall withstand 25,000 impulse cycles at 275 ºF fluid temperature and 75,000 impulse cycles at 225 ºF without evidence of leakage or structural failure (see 4.5.1.6).

3.6.3 Housing and element pressure drop. The maximum pressure drop at rated flow between the inlet and outlet of the filter housing only, shall not exceed 15 psi with a free-flow dummy element installed (see 4.5.1.3). The maximum pressure drop at rated flow across the filter element shall not exceed the value specified in the applicable specification sheet (see 4.5.2.5).

3.6.3.1 Pressure drop differential. The specified filter elements that undergo a second pressure drop test in accordance with 4.5.2.5 shall have a pressure differential between the initial pressure drop values and this pressure drop that shall not exceed ±25 percent and shall not exceed the value specified in the applicable specification sheet (see 4.5.2.14).

3.6.4 Burst pressure. Filter assemblies shall withstand a burst pressure of 250 percent rated pressure (see 4.5.1.7).

3.6.5 Proof pressure. Filter assemblies shall withstand a proof pressure of 150 percent rated pressure at 275 ºF fluid temperature (see 4.5.1.2 and 4.5.1.2.2).

3.6.6 Filter element efficiency. Filter element efficiency shall be as specified in the applicable specification sheet and shall meet the requirements specified therein (see 4.5.2.7).

3.6.6.1 Filter element unloading. Filter elements shall not unload when subjected to intermittent hydraulic pulsing or start up flow conditions as found in typical hydraulic systems (see 4.5.2.11).

3.6.6.2 Filter element integrity. Filter elements shall not be sensitive to variations in contaminant concentration, type or distribution above the filter efficiency rating (see 4.5.2.7).

3.6.6.3 Filter media. Filter elements shall be so designed that under normal handling the filter media shall not be damaged or degraded.

3.6.6.4 Filtering media construction. Filter element filtering media construction shall not be affected by thermal cycling and pressure-flow cycling. Normal operation shall not cause continuous or sporadic release of the media, binder, and adhesive if used and shall not result in degradation of element below rated efficiency level requirements (see 4.5.2.4 and 4.5.2.13).

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3.6.7 Media migration. Filter element assemblies shall be able to withstand vibration frequencies in accordance with 4.5.2.13.3.2. There shall be no media migration traceable to the filter element, its adhesives, or its packaging.

3.6.8 Element collapse. Filter elements shall withstand a differential pressure as specified in the applicable specification sheet (see 4.5.2.10).

3.6.9 Bubble point. All filter elements shall be tested in accordance with 4.5.2.2.1 and values recorded. The initial bubble point values shall be determined as specified in 4.5.2.2.1. All elements submitted to the conformance bubble test shall meet the values established.

3.6.9.1 Initial bubble point. This initial bubble point value is the bubble point established at the beginning of qualification testing (see 4.5.2.2.1). This value will differ for various filtration media by each filter manufacturer. No filter element will be accepted that is below 3 inches water column, though the filter manufacturer may specify a higher value to be used as pass/fail initial bubble point criteria.

3.6.9.2 Minimum bubble point. This bubble point value is defined by successfully conducting the filter conditioning tests and determining the lowest bubble point of the test samples to occur after cold start (see 4.5.2.2.1). This value will differ for various filtration media by each filter manufacturer. This minimum bubble point value shall be identified.

3.6.9.3 Bubble point after performance test. As specified in the qualification test procedure of this specification, a bubble point test shall be conducted before and after each qualification test as specified in table I. The filter element bubble point value shall not be less than the initial bubble point value determined in 3.6.9.1.

3.7 Markings.

3.7.1 Direction of flow. The direction of flow shall be clearly and permanently indicated by at least two arrows marked on opposite sides of the filter body.

3.8 Identification of product.

3.8.1 Filter housing. Each filter housing shall be clearly and permanently identified by steel stamping, electro-etching, applicable laser marking (optional), or by a permanently attached nameplate conforming to MIL-STD-130. The following information shall be provided:

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FILTER, FLUID PRESSURE, HYDRAULIC, (insert pressure rating and micron rating) MICRONS, MAXIMUM TEMPERATURE 275 ºF
M8815/ (insert slash number of specification sheet and size dash number)
Manufacturer's part no.
Manufacturer's serial no.
Manufacturer's name or trademark
Replacement element M8815/ (insert slash number of specification sheet and size dash number)
Part number(s) preformed packing seals
Assembly torque for bowl

3.8.2 Filter element. Each filter element shall be clearly and permanently identified on one end by steel stamping, electro-etching, or by a permanently attached nameplate in accordance with MIL-STD-130. The method of marking shall be in accordance with MIL-STD-130. The marking shall not be placed on a sealing surface; the markings shall not affect the sealing of the element. The following information shall be provided:

M8815/ (insert slash number of specification sheet and size dash number)
Manufacturer's part no.
Manufacturer's serial no. (Cleanable elements only)
Manufacturer's lot no. (Noncleanable elements only)
Manufacturer's name or trademark
Cleanable or non-cleanable
Maximum pressure drop after cleaning (Cleanable elements only; see 4.5.2.10.2.2.1)

3.8.3 Part numbering of interchangeable parts. The manufacturer's part number and drawing number shall be the same.

3.9 Workmanship. All details of workmanship shall be of a sufficiently high grade to ensure proper operation.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in SAE-AS8775 and as specified herein.

4.2.1 Test fluid. Unless otherwise specified, the hydraulic fluid used for all tests shall conform to MIL-PRF-5606, MIL-PRF-83282, or MIL-PRF-87257 as applicable. There shall be no free water in fluid used for testing.

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4.2.1.1 Test fluid cleanliness. For conformance tests and critical qualification tests, the test fluid shall be pre-cleaned to a cleanliness level equal to Class 1 of SAE-AS4059. Differential Particle Count method or Class 1B-F of SAE-AS4059 Cumulative Particle Count may be used as an alternate method.

4.2.1.2 Test fluid filtration. Unless otherwise specified, the test fluid shall be continuously filtered through a 3-micron absolute filter conforming to MIL-PRF-81836 or a filter with equivalent efficiency or higher during testing.

4.2.1.3 Test fluid temperature. Unless otherwise specified, the actual temperature of the test fluid shall be 100 ºF ± 10 ºF.

4.2.2 Ambient temperature. Unless otherwise specified, the tests shall be conducted at a room temperature of 70 ºF to 90 ºF measured within 12 inches of the test sample.

4.2.3 Test air. The air used to pressurize the test apparatus shall be filtered through a 0.45-micron membrane filter (type HA Millipore, or equivalent).

4.2.4 Weight, measurement of.

4.2.4.1 Precision (of weighing). Unless otherwise specified, all weights shall be measured with a precision of one part in one thousand or better.

4.2.4.2 Static electricity, elimination of. A device to eliminate static electricity shall be in proximity to the filter membrane disc, or to the pan on which it rests, whenever any object or material is being weighed to a precision of ±0.1 gram or finer. The age of the device shall not exceed the value stated by the manufacturer as its rated service life.

4.2.5 Test dust, source of supply. All test dust used for tests shall be obtained from the Naval Air Warfare Center Aircraft Division (NAWCAD), (Attention: Code AB24200), Patuxent River, MD. 20670, and shall be the lot of test dust currently in use at NAWCAD.

4.3 Qualification tests.

4.3.1 Filter housing.

4.3.1.1 Test samples. Samples for qualification tests of filter housings shall consist of two specimens of each device upon which qualification is desired. These samples shall undergo the tests outlined in the filter assembly section of table I.

4.3.1.2 Assembling parts. The specimens shall be assembled from parts that conform to manufacturer's drawings.

4.3.1.3 Clearance of moving parts. The manufacturer shall provide calculations showing that adequate clearance of moving parts is provided at -65º and +275 ºF, using the most adverse dimensions. The room temperature reference point shall be 70 ºF.

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4.3.2 Filter elements.

4.3.2.1 Test samples. Qualification of filter elements (cleanable and non-cleanable) shall consist of fourteen elements of each size upon which qualification is desired. Four of these elements will be designated for use as spares, intended to be used if damage is incurred on other elements outside of the test environment. Following the 4.5.2.2.2 sequence determination, two elements shall be tested to one of the five test sequences outlined in the filter element section of table I. Manufacturer's qualification tests shall not have been performed on any samples.

4.3.2.2 Limitation. If the elements are of the type in which repair material is applied to eliminate imperfections, the qualification test elements shall have a minimum of 5 percent of the filtering area repaired. The areas repaired shall contain at least three imperfections whose initial bubble point is 3 inches of water or less.

4.3.3 Tests. The qualification tests shall consist of the tests specified in 4.5 conducted on the applicable specimens in the order specified in table I. Any test event that does not meet applicable test requirements shall be considered a "test failure."

4.4 Conformance inspection. Conformance inspection shall consist of the following:

a. Individual inspection (4.4.1).
b. Sampling tests (4.4.2).

4.4.1 Individual inspection. The individual inspection shall be conducted on all filter housings and filter elements in the order specified in table II. Any filter housing or filter element containing a defect will be rejected.

4.4.2 Sampling tests. A sample shall be selected from each inspection lot (see 4.4.3) in accordance with ANSI/ASQ Z1.4 using special inspection level S-1 with no rejects allowed. The sample filter assemblies and filter elements shall be unpacked and the sampling tests shall be conducted on each filter assembly and filter element in the order specified in table II. For filter elements one half of the samples shall be subjected to the pressure buildup and collapse test and one half of the samples shall be subjected to the degree of filtration test. When the total number of samples is an odd number, the odd sample shall be subjected to the pressure buildup and collapse test.

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TABLE I. Qualification tests.

Filter assembly (Housing and element)	Requirement Reference	Verification Reference
Examination of product	3.5.1	4.5.1.1
Proof pressure	3.6.5	4.5.1.2
Housing pressure drop	3.6.3	4.5.1.3
Protective devices	3.5.6	4.5.1.4
Extreme temperature	3.5.1	4.5.1.5
Proof pressure	3.6.5	4.5.1.2
Media and abrasion migration	3.6.7	4.5.2.13
Impulse	3.6.2	4.5.1.6
Protective devices 1/	3.5.6	4.5.1.4
Burst pressure	3.6.4	4.5.1.7
Filter Element Test Sequence No. 1 (lowest bubble point)	Requirement Reference	Verification Reference
Examination of product	3.5.1	4.5.2.1
Bubble point	3.6.9	4.5.2.2.1
Immersion and thermal cycling	3.6.6.4	4.5.2.4
Initial pressure drop	3.6.3	4.5.2.5
Bubble point	3.6.9	4.5.2.2.1
Cold start		4.5.2.6
Bubble point	3.6.9	4.5.2.2.2
Maximum particle passed		4.5.2.8
Bubble point	3.6.9	4.5.2.2.3
Flow fatigue		4.5.2.9
Bubble point	3.6.9	4.5.2.2.3
Filter Element Test Sequence No. 2 (highest bubble point)	Requirement Reference	Verification Reference
Examination of product	3.5.1	4.5.2.1
Bubble point	3.6.9	4.5.2.2.1
Immersion and thermal cycling	3.6.6.4	4.5.2.4
Initial pressure drop	3.6.3	4.5.2.5
Bubble point	3.6.9	4.5.2.2.1
Cold start		4.5.2.6
Bubble point	3.6.9	4.5.2.2.2
Pressure buildup and collapse	3.6.8	4.5.2.10
Bubble point	3.6.9	4.5.2.2.3
Relief valve operation (if applicable)	3.5.6.2	4.5.2.12
Filter Element Test Sequence No. 3 (2nd lowest bubble point)	Requirement Reference	Verification Reference
Examination of product	3.5.1	4.5.2.1
Bubble point	3.6.9	4.5.2.2.1
Immersion and thermal cycling	3.6.6.4	4.5.2.4
Initial pressure drop	3.6.3	4.5.2.5
Bubble point	3.6.9	4.5.2.2.1
Cold start		4.5.2.6

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TABLE I. Qualification tests - Continued.

Filter Element Test Sequence No. 3 (2nd lowest bubble point)	Requirement Reference	Verification Reference
Bubble point	3.6.9	4.5.2.2.2
Degree of filtration	3.6.6	4.5.2.7
Bubble point	3.6.9	4.5.2.2.3
Filter element unloading	3.6.6.1	4.5.2.11
Filter Element Test Sequence No. 4	Requirement Reference	Verification Reference
Examination of product	3.5.1	4.5.2.1
Bubble point	3.6.9	4.5.2.2.1
Immersion and thermal cycling	3.6.6.4	4.5.2.4
Initial pressure drop	3.6.3	4.5.2.5
Bubble point	3.6.9	4.5.2.2.1
Cold start		4.5.2.6
Bubble point	3.6.9	4.5.2.2.2
Filter element flush		4.5.2.3
Media and abrasion migration	3.6.7	4.5.2.13
Bubble point	3.6.9	4.5.2.2.3
Filter Element Test Sequence No. 5	Requirement Reference	Verification Reference
Examination of product	3.5.1	4.5.2.1
Bubble point	3.6.9	4.5.2.2.1
Immersion and thermal cycling	3.6.6.4	4.5.2.4
Initial pressure drop	3.6.3	4.5.2.5
Bubble point	3.6.9	4.5.2.2.1
Cold start		4.5.2.6
Bubble point	3.6.9	4.5.2.2.2
Visual examination		4.5.2.14.1
Pressure drop	3.6.3.1	4.5.2.14.2

1/ Make functional checks only one time, omitting any required repeated cycling.

4.4.2.1 Cleanliness. To ensure that the filter elements and housing are being cleaned properly, and to ensure that the parts are not contaminated during packaging, sample filter elements or assemblies shall be unpacked and tested for cleanliness as specified in 4.4.2.1.1 and 4.4.2.1.2. Filter elements shall be installed in a pre-cleaned housing.

4.4.2.1.1 Vibration for cleanliness determination. The filter assembly, filled with the test fluid, shall be vibrated for 15 minutes at a frequency of 120 ±10 cycle per second (cps) and a double amplitude of 0.014 inch in a direction parallel to the bowl axis of the assembly.

4.4.2.1.2 Degree of cleanliness. The filter assembly shall be transferred to a stand and flushed with 2,000 milliliters of pre-filtered hydraulic test fluid at rated flow, and the effluent collected. The contaminant level of the effluent sample shall be determined in accordance with SAE-ARP785. The contaminant value obtained shall not exceed 1.75 milligrams maximum.

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TABLE II. Conformance tests.

Individual Inspections	Requirement Reference	Verification Reference
Filter Housing		
Examination of product	3.5.1	4.5.1.1.1
Proof pressure	3.6.5	4.5.1.2.2
Protective devices		
Relief valve	3.5.6.1	4.5.1.4.1.2
Differential pressure indicator	3.5.6.3	4.5.1.4.2.5
Automatic shutoff	3.5.6.4	4.5.1.4.3.1.2
Filter elements only		
Examination of product	3.5.1	4.5.2.1.2
Bubble point (Conformance)	3.6.9	4.5.2.2.4
Filter Element pressure drop (cleanable elements)	3.6.3	4.5.2.10.3
Filter element relief valve (if applicable)	3.5.6.2	4.5.2.12.2
Sampling tests	Requirement Reference	Reference
Filter assemblies and filter elements		
Cleanliness		4.4.2.1
Filter Elements		
Pressure buildup and collapse (One loading only to applicable pressure drop specified in Table I of specification sheet – no samples taken)	3.6.8	4.5.2.10
Degree of filtration (Filter element efficiency)	3.6.6	4.5.2.7
Filter element unloading	3.6.6.1	4.5.2.11

4.4.3 Inspection lot. For purposes of conformance inspection, a lot shall be defined as all filter assemblies manufactured under essentially the same conditions and offered for acceptance at one time, or all filter elements manufactured from one convoluted setup.

4.5 Test methods.

4.5.1 Filter assembly (housing and element).

4.5.1.1 Examination of product (qualification and conformance). The filter housing to be tested shall be carefully examined to determine conformance to the requirement of this specification for design, weight, workmanship, marking, conformance to applicable AN or MS standard, government and manufacturer's drawings, specification sheets and standards for any visible defects.

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4.5.1.2 Proof pressure.

4.5.1.2.1 Proof pressure (qualification). Filter assemblies shall withstand proof pressure as specified herein without evidence of permanent deformation, malfunction, or external leakage. The filter assembly shall be filled with test fluid and maintained at 275 ºF for 72 hours. Proof pressure of 150 percent rated pressure shall then be applied twice while at 275 ºF and held for 2 minutes at each application. The pressure shall be reduced to zero between applications. The activity responsible for qualification shall have the prerogative to conduct additional tests to validate compliance of the filter assemblies with the longevity requirements specified in 3.5.1.

4.5.1.2.2 Proof pressure (conformance). Proof pressure of 150 percent rated pressure shall be applied twice at room temperature and held for 2 minutes at each application. The pressure shall be reduced to zero between pressure applications. There shall be no evidence of permanent deformation, malfunction, or external leakage.

4.5.1.3 Housing pressure drop. This test shall be conducted with a free-flow dummy element installed to hold open the automatic shut off. The free-flow dummy element shall be constructed in accordance with the applicable filter element specification sheet with the X dimension being held to the minimum diameter shown (see MIL-DTL-8815/3). The filter manufacturer shall have the prerogative to furnish this test dummy with filtration media representing minimum pressure drop. With rated flow applied to the filter housing, the pressure drop shall not exceed 15 psi.

4.5.1.4 Protective devices. A means of stopping flow through the filter element port shall be installed for this test.

4.5.1.4.1 Relief valve operation.

4.5.1.4.1.1 Relief valve (qualification). Style A filter housings shall be tested for relief valve operation. By means of a power-driven pump, pressure shall be applied to the inlet port of the filter housing, beginning with a pressure of 70 psi and increasing in increments of 10 psi or less until cracking pressure of the relief valve in the filter housing being tested is reached. At each increment, the pressure shall be maintained constant for 5 minutes. At each pressure increment, time shall not be considered “in” or leakage noted until the beginning of the third minute. The leakage rate during the last 3 minutes of each pressure increment shall be noted. The pressure at which the leakage rate through the relief valve amounts to 8 cc (approximately 160 drops) per minute shall be considered as cracking pressure. Cracking pressure shall occur at 100 ±10 psi or as required in accordance with the applicable specification sheet. Leakage up to 80 psi shall not exceed 4 cc (approximately 80 drops) per minute. Pressure shall be increased until rated flow through the relief valve is obtained. Rated flow shall occur at a pressure differential between inlet and outlet ports not exceeding 160 psi. The pressure shall be reduced by 10 psi decrements as above until leakage through the relief valve does not exceed the rate of 4 cc per minute. The resulting pressure shall be considered the relief valve reseating pressure and shall occur at a pressure of 65 psi or above. The flow through the relief valve shall be raised to 150 percent of rated flow. The differential pressure across the relief valve shall not exceed 240 psi.

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4.5.1.4.1.1.1 Relief valve contamination tolerance test. Style A filter housings shall repeat the qualification test of 4.5.1.4.1.1, with the fluid contaminated to a level of 100 mg/L with A.C. fine test dust (see 6.5). The cracking, full flow and reseat cycle shall be applied for a total of 100 cycles. At the end of each cycle, the flow shall be reduced to zero prior to commencing the next cycle. Malfunction or relief valve damage shall constitute failure of this test.

4.5.1.4.1.2 Relief valve (conformance). The sampled filter housing relief valve shall be tested as follows:

a. Style A filter housing shall be tested for relief valve operation. By means of a power-driven pump, pressure shall be applied to the inlet port of the filter housing. At each pressure increment, the pressure shall be maintained constant for 3 minutes, except that for the first increments, adequate time shall be allowed to permit all cavities to fill with fluid and steady state conditions to be obtained.

b. The pressure at the inlet port shall initially be raised to 90 psi. If steady state leakage through the relief valve is less than 4 cc (approximately 80 drops) per minute, the test shall be continued with item (d).

c. If leakage is between 4 cc (approximately 80 drops) and 8 cc (approximately 160 drops) per minute, the inlet pressure shall be dropped below 65 psi and then raised to 80 psi. Leakage at 80 psi shall not exceed 4 cc per minute.

d. The inlet pressure shall then be raised to 110 psi. Flow through the relief valve shall exceed 8 cc per minute. The inlet pressure shall be further increased to produce a flow of at least 25 percent of rated flow through the relief valve. This pressure shall be below 150 psi. The inlet pressure shall then be reduced to 65 psi. Leakage through the relief valve shall be less than 4 cc per minute.

4.5.1.4.1.3 Relief valve cycling test in style A housings. The filter housing relief valve shall be subjected to a cycling test as follows: With the outlet port open, and an effective means for stopping flow through the filter element installed, the relief valve shall be cycled 2,000 times from zero to 150 percent of rated flow and back at 100 ºF ±10 ºF. There shall be no malfunction of any of the relief valve parts during this test. Upon completion of the cycling test, the relief valve shall meet all the requirements of the relief valve operating test with the exception that the leakage up to 80 psi shall not exceed 5 cc (approximately 100 drops) per minute.

4.5.1.4.2 Differential pressure indicator operation. For all tests where indicator operation is required, the ambient and fluid temperature shall be at 120 ºF minimum unless otherwise specified to ensure the release of the thermal lockout mechanism. Alternate test procedures must be approved by the activity responsible for qualification (see 6.3) prior to implementation. For shock test (see 4.5.1.4.2.2) and media and abrasion migration test (see 4.5.2.13.3.2), the thermal lockout mechanism shall be inactivated or the test conducted with an ambient and fluid temperature of 120 ºF minimum.

4.5.1.4.2.1 Pressure actuation (qualification).

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4.5.1.4.2.1.1 Mechanical differential pressure indicator pressure actuation. Styles A and B filter housings shall be tested for differential pressure indicator operation with the element port plugged and outlet port vented to atmosphere. By means of a power-driven pump, the pressure shall be slowly increased at the inlet port of the filter housing until the pressure indicator actuates fully and locks. This shall be considered the indicator operating pressure. The indicator operating pressure shall occur at 70 ±10 psid. The pressure shall then be raised to 150 percent rated pressure and dropped to zero psi. The indicator shall stay locked in the extended position throughout the pressure range of zero to 150 percent rated pressure. The indicator shall then be manually reset. For style A filter housings, the relief valve shall be blocked. This test shall be repeated at a fluid and ambient temperature of 80 ºF. No actuation shall occur.

4.5.1.4.2.1.2 Electrical differential pressure indicator pressure actuation. With the ambient and fluid temperature stabilized at 80 ºF maximum and 0 psi applied, the warning light shall demonstrate "push to test" functionality (or similar) and provide positive feedback of normal operation. A pressure of 80 psi shall then be applied by a power-driven pump to the inlet port of the filter housing, with the outlet port vented. The warning light shall illuminate GREEN. If a temperature sensing device is incorporated, there shall be no indication of high temperature. Release the pressure. Stabilize the ambient and fluid temperature at 120 ºF minimum and apply a pressure of 40 psi. The warning light shall illuminate GREEN. Increase the pressure to 60 psi and the warning light shall illuminate YELLOW during this pressure rise from 40 to 60 psi. Increase the pressure to 80 psi and the warning light shall illuminate RED during this pressure rise from 60 to 80 psi. Reduce pressure to 0 psi and the warning light shall remain illuminated RED. Reset warning light and light shall illuminate GREEN. If a temperature sensing device is incorporated, apply a pressure of 40 psi to the inlet port of the filter housing and increase fluid temperature to 270 ºF minimum. Allow the temperature to stabilize for 5 minutes. Reduce the fluid temperature to ambient conditions and reduce inlet pressure to 0 psi. Once stabilized at ambient temperatures, the device shall illuminate RED.

4.5.1.4.2.1.2.1 Electrical differential pressure indicator qualification. The electrical differential pressure indicator shall meet all SAE-AS8775 requirements for Class C electrically operated components.

4.5.1.4.2.1.3 Low system pressure cycling. With the filter element port plugged and the outlet port vented to atmosphere, the pressure shall be increased at a rate of 20 psi per minute at the inlet port of the filter housing, until the pressure indicator actuates fully or illuminates RED (for electrical differential pressure indicators). This shall be considered the indicator operating pressure and shall occur at 70 ±10 psid or as required in accordance with the applicable specification sheet. For electrical differential pressure indicators, the pressure indicator shall illuminate YELLOW at 40 to 60 psi. The inlet pressure shall then be reduced to zero and the indicator manually reset. The pressure shall then be cycled 1,000 times at any practical cycling rate from zero to 80 psi and back to zero psi. The indicator shall actuate and shall be reset at the end of each cycling and shall occur at 70 ±10 psid or as required in accordance with the applicable specification sheet.

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4.5.1.4.2.1.4 High system pressure cycling. The filter element port shall be blocked with a plug containing an orifice designed to provide rated flow at 80 psi differential pressure across the orifice. The outlet port of the filter housing shall be throttled to provide rated flow through the orifice at rated pressure to the inlet. The flow shall be cycled from zero to rated flow and the differential pressure at which the indicator actuates fully or illuminates RED (for electrical differential pressure indicators) shall be noted. This shall occur at a differential pressure of 70 ± 10 psid or as required in accordance with the applicable specification sheet. The inlet pressure shall then be raised to rated pressure and dropped to zero psi. After actuation, the indicator shall remain in the mechanically extended position or shall continue to illuminate RED. The indicator shall then be manually reset. This cycle shall be repeated 1,000 times. The differential pressure indicator operating pressure at rated pressure to the inlet shall then be rechecked and shall occur at 70 ± 10 psid or as required in accordance with the applicable specification sheet.

4.5.1.4.2.1.5 Flow surge. Styles A and B filter housings shall be tested for inadvertent differential pressure indicator operation due to a temporary flow surge. The filter element port shall be blocked. With a power-driven pump and a quick opening valve, the pressure shall be rapidly raised from 0 to 80 psi. The differential pressure indicator shall not actuate or illuminate during the first 0.1 second of pressure application. The differential pressure indicator shall actuate or illuminate within 1 second from the start of the pressure application. The test shall be repeated at 200, 300, 400, and 500 psi and the differential pressure indicator shall actuate or illuminate within the specified 0.1 to 1 second time range. For Style A filter housings, the relief valve shall be blocked for the foregoing test.

4.5.1.4.2.2 Shock. Styles A and B filter assemblies shall be subjected to 12 impact shocks of 20g. Each shock impulse shall have time duration of 10 ±1 milliseconds. The maximum g shall be reached in approximately 5 milliseconds. For mechanically actuated devices, the shocks shall be applied in each of the three principal filter assembly axes with the differential pressure indicator in an extended and retracted position. The differential pressure indicator shall remain in its original extended or retracted position during the tests. For electronically actuated devices, the shocks shall be applied in each of the three principal filter assembly axes with the differential pressure indicator set to both normal (GREEN) and loaded (RED) illuminations. The differential pressure indicator shall not change illumination during the test. The shock test shall not have any detrimental effect on the differential pressure indicator, relief valve or automatic shut off.

4.5.1.4.2.3 Dust test. Styles A and B filter assemblies shall be subjected to dust test procedure in accordance with MIL-STD-810, method 510.7. The indicator operating pressure shall be checked in accordance with 4.5.1.4.2.1.1 following this test and shall occur at 70 ±10 psid or as required in accordance with the applicable specification sheet.

4.5.1.4.2.4 Salt fog. Styles A and B filter assemblies shall be subjected to salt fog test procedure in accordance with MIL-STD-810, method 509.7. The differential pressure indicator shall be checked in accordance with 4.5.1.4.2.1 before removing any salt deposits and again after rinsing with water. The indicator shall actuate at 70 ±10 psid or as required in accordance with the applicable specification sheet.

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4.5.1.4.2.5 Pressure actuation (conformance). Styles A and B filter housings shall be tested for differential pressure indicator operations with the element port plugged.

4.5.1.4.2.5.1 Mechanical differential pressure indicator. With the ambient and fluid temperature stabilized at 80 ºF maximum, a pressure of 80 psi shall be applied by a power-driven pump to the inlet port of the filter, with the outlet port vented. The indicator shall not actuate. Release the pressure. Stabilize the ambient and fluid temperature at 120 ºF minimum and apply a pressure of 59 psi. The indicator shall not actuate. Increase the pressure to 80 psi and the indicator shall actuate during this pressure rise from 60 to 80 psi. Reduce pressure to 0 psi and the indicator shall remain in the actuated position. Depress button manually and button shall remain depressed.

4.5.1.4.2.5.2 Electrical differential pressure indicator. With the ambient and fluid temperature stabilized at 80 ºF maximum and 0 psi applied, the warning light shall demonstrate "push to test" functionality (or similar) and provide positive feedback of normal operation. A pressure of 80 psi shall then be applied by a power-driven pump to the inlet port of the filter, with the outlet port vented. The warning light shall illuminate GREEN. If a temperature sensing device is incorporated, there shall be no indication of high temperature. Release the pressure. Stabilize the ambient and fluid temperature at 120 ºF minimum and apply a pressure of 40 psi. The warning light shall illuminate GREEN. Increase the pressure to 60 psi and the warning light shall illuminate YELLOW during this pressure rise from 40 to 60 psi. Increase the pressure to 80 psi and the warning light shall illuminate RED during this pressure rise from 60 to 80 psi. Reduce pressure to 0 psi and the warning shall remain illuminated RED. Reset warning light and light shall illuminate GREEN. If a temperature sensing device is incorporated, apply a pressure of 40 psi to the inlet port of the filter and increase fluid temperature to 270 ºF minimum. Allow the temperature to stabilize for 5 minutes. Reduce the fluid temperature to ambient conditions and reduce inlet pressure to 0 psi. Once stabilized at ambient temperatures, the device shall provide indication of high temperature.

4.5.1.4.3 Automatic shutoff.

4.5.1.4.3.1 Leakage. The following tests shall be performed on styles A and B filter housings.

4.5.1.4.3.1.1 Automatic shutoff (qualification). A pressure of 5 psi shall be applied simultaneously to both ports of the filter assembly. The bowl and filter element shall be disengaged, removed, and the excess fluid shall be removed from the lower surface of the filter housing head. The leakage measurement period shall be 30 minutes in duration and shall begin 2 minutes after removal of excess fluid. The foregoing test shall be repeated using a pressure of 200 psi. There shall be no external leakage, other than a slight wetting insufficient to form a drop.

4.5.1.4.3.1.2 Automatic shutoff (conformance). A pressure of 5 psi shall be applied simultaneously to both ports of the filter assembly. The bowl and filter element shall be disengaged, removed, and the excess fluid removed from the lower surface of the filter housing head. The leakage measurement period shall be 2 minutes in duration after removal of excess

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fluid. The foregoing test shall be repeated, using a pressure of 200 psi. There shall be no measurable leakage during this test.

4.5.1.4.3.2 Cycling. With a test fluid pressure of 200 psi applied simultaneously to both ports of the filter assembly, the bowl and filter element shall be disengaged, removed, and reinstalled a total of 500 times. No malfunction of the shutoff device shall occur as a result of this cycling. At the conclusion of the cycling, the shutoff device shall pass the leakage test (see 4.5.1.4.3.1). The filter head, bowl, and filter element shall be inspected and show no signs of excessive wear. Filter bowl and filter element seals may be replaced periodically for ordinary wear, and shall not exhibit damage due to cutting or pinching.

4.5.1.4.3.3 Air inclusion. Calculations shall be made indicating the maximum possible air inclusion, assuming that the filter bowl is completely filled with fluid prior to assembly. The maximum calculated air inclusion shall not exceed 2 cc.

4.5.1.4.3.4 Flow assurance. With the filter element removed and the filter bowl installed, pressure shall be applied to the flow inlet port of the filter assembly by means of a power-driven pump, until rated flow occurs from the outlet port. The differential pressure through the filter housing shall not exceed 200 psi.

4.5.1.5 Extreme temperature. The following tests shall be performed on styles A and B filter housings, as applicable.

4.5.1.5.1 Style A. With the filter element port blocked, style A filter housings shall be subjected to a temperature of -65 ºF to -70 ºF for 24 hours. During this period, pressure applied shall not exceed 10 psi. At the end of this period, operation of the relief valve shall be checked for cracking and reseat only. Proof pressure shall then be applied simultaneously at both ports for 2 minutes. The temperature of the hydraulic fluid used for these tests shall be -65 ºF to -70 ºF. Within 20 minutes, the filter housing shall be removed from the cold box and placed in an ambient temperature of 275 ºF. A pressure of 80 psi shall be applied to the inlet port with the outlet port vented to atmosphere. The fluid temperature at which the differential pressure indicator actuates shall be measured and shall be 100 ºF ±15 ºF. The indicator operating pressure and relief valve cracking and reseat pressure shall be checked at 5, 10, and 60 minutes, and 24 hours following the first indicator actuation. Indicator operating and relief valve cracking and reseat pressures shall be within the ranges specified in 4.5.1.4.1 and 4.5.1.4.2.1.

4.5.1.5.2 Style B. With the element port blocked, style B filter housings shall be subjected to a temperature of -65 ºF to -70 ºF for 24 hours. During this period, pressure applied shall not exceed 10 psi. At the end of this period, a pressure of 80 psi shall be applied to the inlet port of the unit and held for 10 minutes, with the outlet port vented to atmosphere. Proof pressure shall then be applied to the inlet port for 2 minutes. The temperature of the hydraulic fluid used for these tests shall be -65 ºF to -70 ºF. Within 20 minutes, the filter housing shall be removed from the cold box and placed in an ambient temperature of 275 ºF. A pressure of 80 psi shall be applied to the inlet port of the unit with the outlet port vented to atmosphere. The fluid temperature at which the differential pressure indicator actuates shall be measured and shall be 100 ºF ±15 ºF. The indicator operating pressure shall be checked at 5, 10 and 60 minutes, and 24

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hours after the first indicator actuation. Indicator operating pressure shall be within the range specified in 4.5.1.4.2.1.

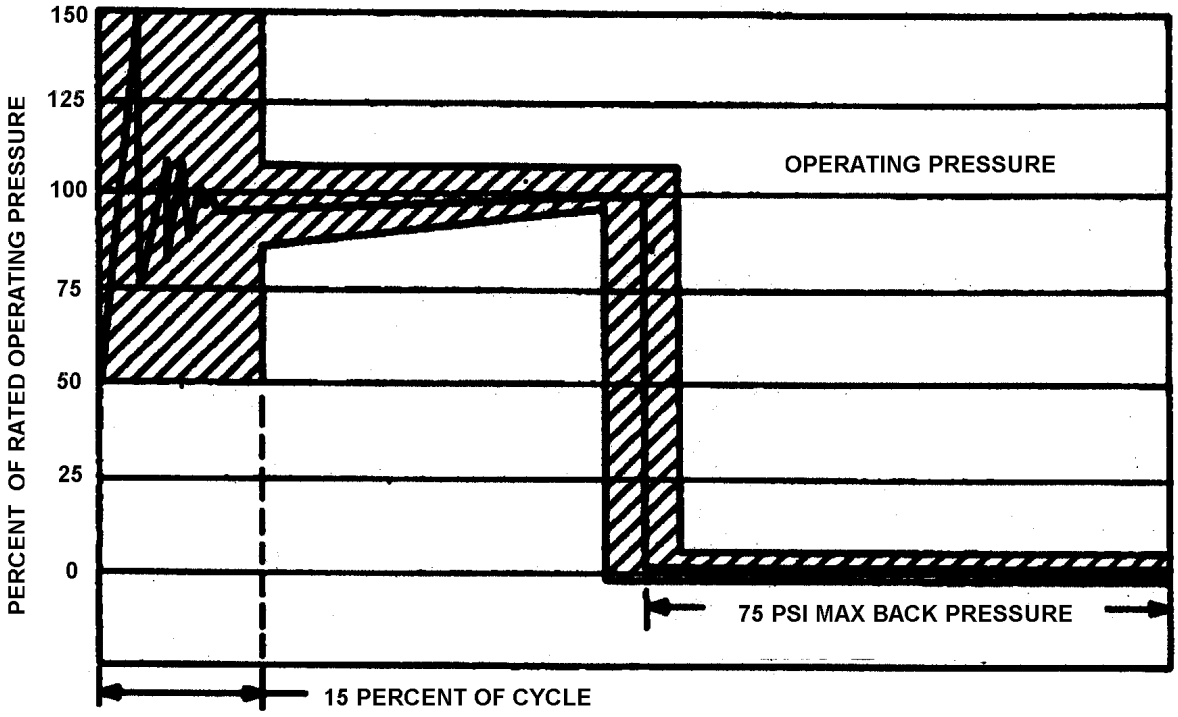
4.5.1.6 Impulse. Filter assemblies shall be subjected to 100,000 impulse cycles, 25,000 at a fluid temperature of 275 ºF and 75,000 at a fluid temperature of 225 ºF. Each impulse cycle shall consist of a pressure rise from zero to 150 percent of the rated pressure and drop to zero. During each pressure increase, a peak surge pressure of 1.50 ±0.07 times (1.43 to 1.57) the rated pressure, as shown by an oscillograph, shall be obtained. Cycling shall be performed at a rate of 300 cycles per minute (cpm) maximum. There shall be no evidence of external leakage or structural failure during the performance of this test. See figure 1 for impulse curve. The differential pressure indicator shall not actuate or illuminate (for electronic differential pressure indicator) during the impulse test. The actual impulse pattern shall be recorded and reported. Following this test, all protective devices shall be re-tested for correct functional operation.

4.5.1.7 Burst pressure. Pressure shall be applied at a maximum rate of increase of 25,000 psi per minute until the specified burst pressure of 250 percent of rated pressure is reached. Filter housings shall show no leakage in the form of drops or rupture of internal or external parts at this pressure when held for 2 minutes. The pressure may be increased above that specified during qualification tests in order to secure data on actual burst pressure. The test shall be conducted at 275 ºF after a 5-hour soak at 275 ºF with the filter housing filled with test fluid.

4.5.2 Filter elements.

4.5.2.1 Examination of product (qualification and conformance). All filter elements to be tested (14 for qualification or sample for conformance) shall be carefully examined to determine conformance to the requirement of this specification for design, weight, workmanship, marking, conformance to applicable AN or MS standard, government and manufacturer's drawings, specifications sheets and standards for any visible defects.

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NOTE:

The curve shown above is the approximate pressure-time cycle determined to be of proper severity for impulse testing. Although it is mandatory only that pressure peak rises to 150 percent of the operating pressure at some point prior to leveling off at rated pressure, it is considered desirable that the pressure-time curve be confined to the shaded area indicated. One very desirable benefit to be gained in this manner is that results of tests performed on different test machines will be more nearly comparable.

Initial rate of rise pressure application to be 200,000-300,000 psi per second.

FIGURE 1. Impulse curve.

4.5.2.2 Bubble point.

4.5.2.2.1 Bubble point qualification (initial bubble point). Each filter element shall be tested to determine the initial bubble point. The filter element, containing no fluid, shall be installed in a setup similar to figure 2. The fluid level shall be maintained at approximately 0.5 inch above the top of the filter element. The air pressure, as indicated in inches of water on the manometer, shall be slowly raised by small increments. The filter element shall be rotated 360 degrees about its longitudinal axis at each increment of air pressure so that the entire filter area can be observed for the appearance of the first bubble. The area of greatest porosity is determined by observing the first bubble on the surface of the filter element; and the manometer reading in inches of water at which this bubble emits from the filter element shall be recorded. This test shall be accomplished within a period of 10 minutes after immersion in the fixture. The fluid used shall be Proprietary Solvent #3 (U.S. Industrial Chemicals), or equivalent, at 70 ºF ±5 ºF filtered

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through 0.45-micron membrane Millipore filter, or equivalent. No filter element will be accepted that is below 3 inches water column or as specified by the filter manufacturer, whichever value is higher.

4.5.2.2.2 Bubble point (sequence determination and minimum bubble point). Following pressure drop, each of the fourteen (14) filter elements shall be tested to determine bubble point in accordance with 4.5.2.2.1. Bubble point values shall be recorded. The lowest bubble point of the fourteen qualification elements shall be considered the minimum bubble point. The minimum bubble point shall be recorded on any released drawings/documentation associated with this filter element. The outcome of this test will also determine which test sequence each filter element will follow. The four (4) elements with the highest bubble point values shall be designated as spare units and set aside. The remaining ten (10) elements shall be ranked accordingly and will follow the remaining test procedures, as outlined in table I. The filter element rankings and sequences are as follows: the two elements (2 articles) with the lowest bubble point rankings shall perform the maximum particle passed and flow fatigue tests. The two elements with the third and fourth lowest bubble point ranking (2 articles) shall perform the degree of filtration tests. The two elements with the fifth and sixth lowest bubble point ranking (2 articles) shall perform the media and abrasion migration tests. The two elements with the highest bubble point rankings (2 articles) shall perform the pressure buildup and collapse and relief valve operation (if applicable) tests. The remaining elements (2 articles) shall undergo the pressure drop tests.

4.5.2.2.3 Bubble point after performance test. Where specified in table I, a bubble point test shall be conducted in accordance with 4.5.2.2.1. The bubble point shall not be less than the minimum bubble point, as determined in 4.5.2.2.2.

4.5.2.2.4 Bubble point (conformance). The filter element, containing no fluid, shall be installed in a setup similar to figure 2. The fluid level shall be maintained at approximately ½ inch above the top of the filter element. The air pressure shall be preset to 0.1inch of water less than the minimum bubble point. The filter element shall be rotated 360 degrees about its longitudinal axis and the entire filter area scanned for the appearance of any bubbles. There shall be no bubbles emerging from the filter. The fluid used shall be Proprietary Solvent #3, or equivalent, at 70 ºF ±5 ºF filtered through a 0.45-micron membrane Millipore filter, or equivalent. Since the minimum bubble point value will differ for various filtration media, the value shall be obtained from the qualification approval letter for the particular filter element under test.

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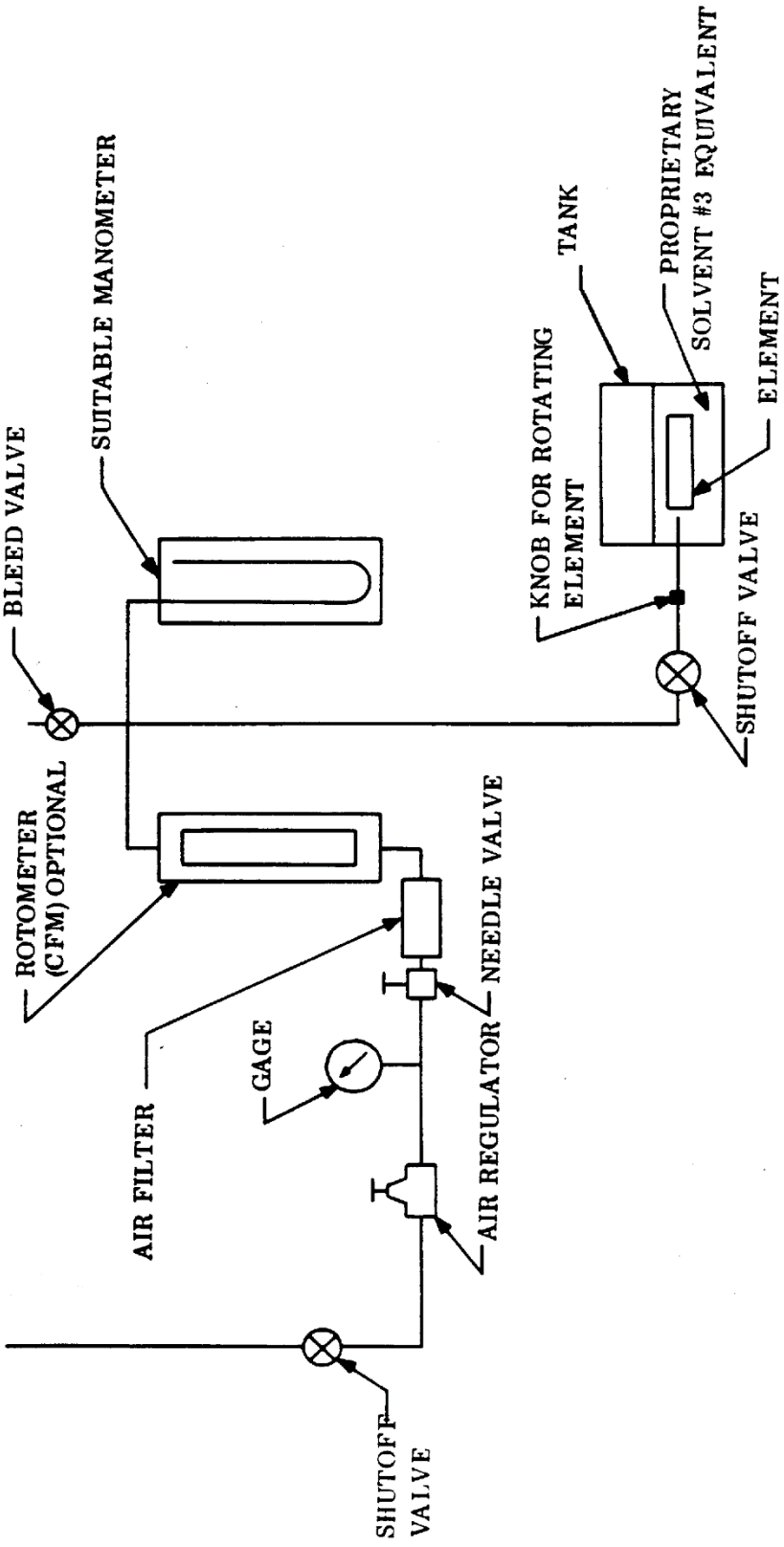


FIGURE 2. Typical schematic diagram for air bubble test.

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4.5.2.3 Filter element flush. The test fluid in the figure 3 system shall be pre-cleaned in accordance with 4.2.1.1. The filter element shall be installed on figure 3 system and flushed at rated flow for 1 hour. The cleanup filter shall not be bypassed while flushing.

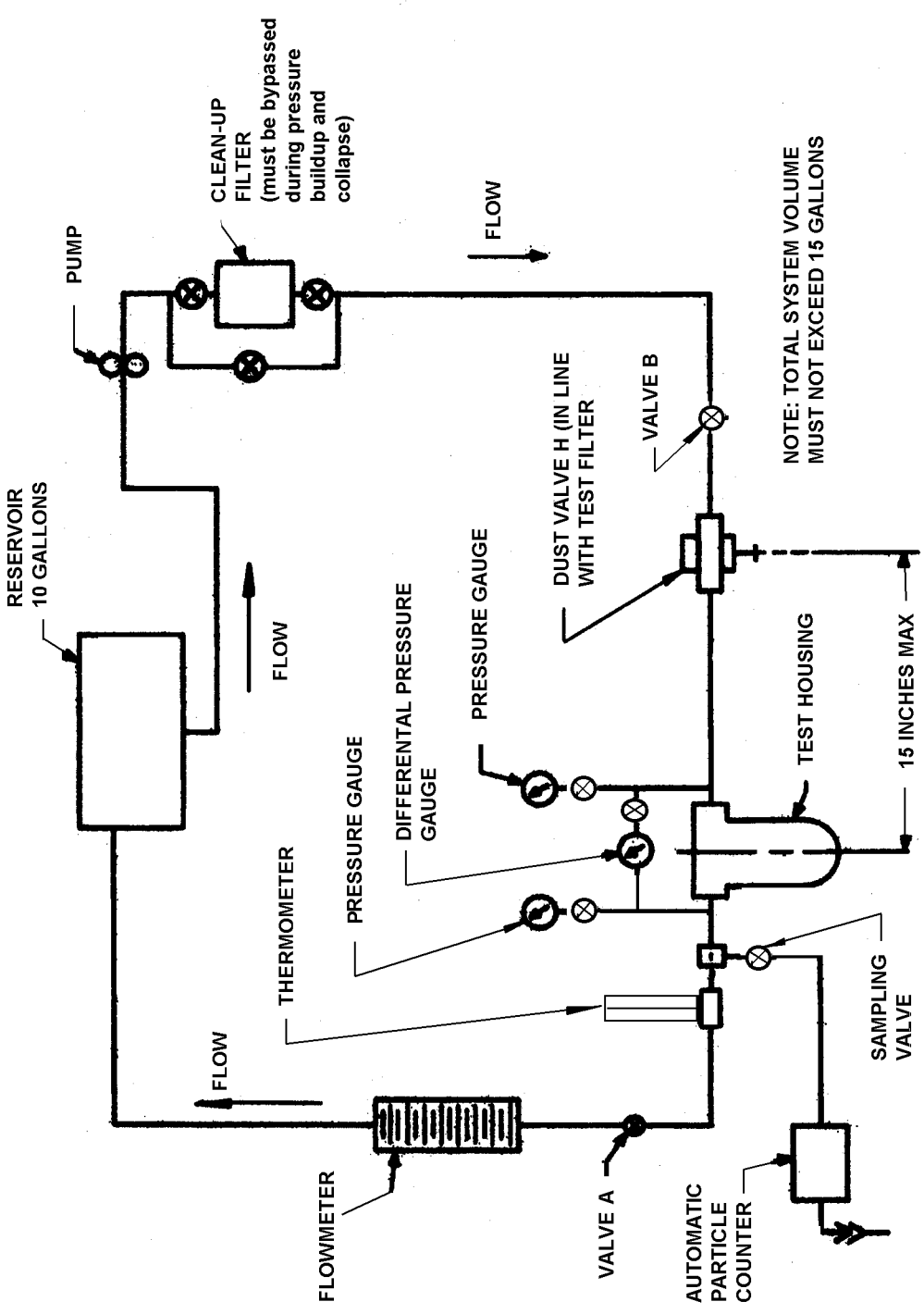


FIGURE 3. Typical setup for determining pressure buildup and collapse pressure characteristics.

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4.5.2.4 Immersion and thermal cycling. Filter elements shall be immersed in hydraulic fluid (see 4.2.1) for a period of 24 hours at a fluid temperature of 275 ºF and 24 hours at a fluid temperature of -65 ºF in two separate environmental chambers. This cycle shall then be repeated two more times for a total period of 144 hours (72 hours at a fluid temperature of 275 ºF and 72 hours at a fluid temperature of -65 ºF). Transfer between hot and cold chambers shall not exceed 10 minutes. All internal parts shall be in contact with the test fluid during this period. After immersion, the filter shall remain in this fluid at room temperature until further tests are conducted.

4.5.2.5 Initial pressure drop. The filter elements shall undergo an initial differential pressure test measuring the differential pressure upstream and downstream of the filter under test using a differential pressure transducer or two pressure transducer gauges as specified in ISO 3968, at 100 °F ± 5°F. The maximum pressure drop across the element shall be as specified in the applicable specification sheet. For example, the maximum pressure drop across the element at 6 gallons per minute (gpm) flow rate shall not be greater than 10 psid. The pressure drop shall be recorded for each element. Note that while the initial pressure drop values are recorded for each filter element, only two (2) elements will undergo a secondary pressure drop sequence following cold start, as determined below.

4.5.2.6 Cold start. Filter elements shall be installed in a housing filled with oil and subjected to a temperature of -65º ±5 ºF for four hours in a test setup similar to figure 4. The elements shall then be subjected to 10 flow cycles at a differential pressure as specified in the applicable specification sheet across the element with fluid held at -65 ºF ±5 ºF or lower. The relief valve (where applicable) shall be blocked. Each flow cycle shall be 15 ±1 seconds in duration. See figure 5 for cold start cycle wave shape. A flow cycle-pressure time trace shall be prepared. There shall be no evidence of filter media damage for each element tested as evidenced by visual examination and satisfactory completion of subsequent tests listed in table I. The test fluid used for this test shall be in accordance with MIL-PRF-5606 or MIL-PRF-87257, which have similar viscosity at -65 ºF.

4.5.2.6.1 Fluid analysis. For filter element test sequence no. 4 during the first and tenth cycle of the above test, 500 mL of test fluid shall be drawn down stream of the test filter. Prior to drawing samples, a volume of fluid equal to the static fluid volume in the sampling line, fittings and valve shall be drawn and discarded. The fluid shall be analyzed in accordance with SAE-ARP785 and 4.5.2.13.3.4. The weight of contaminant collected for each sample shall be multiplied by four and shall not exceed the migration analysis value listed in table 1 of the applicable specification sheet.

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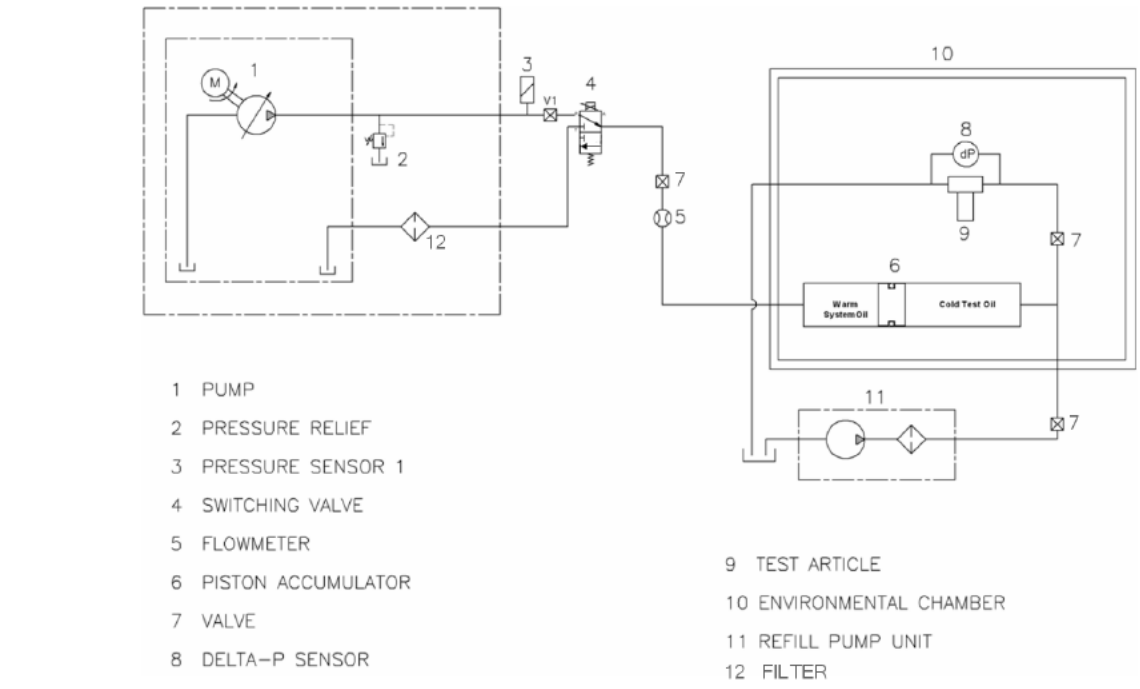


FIGURE 4. Typical schematic diagram for cold start test.

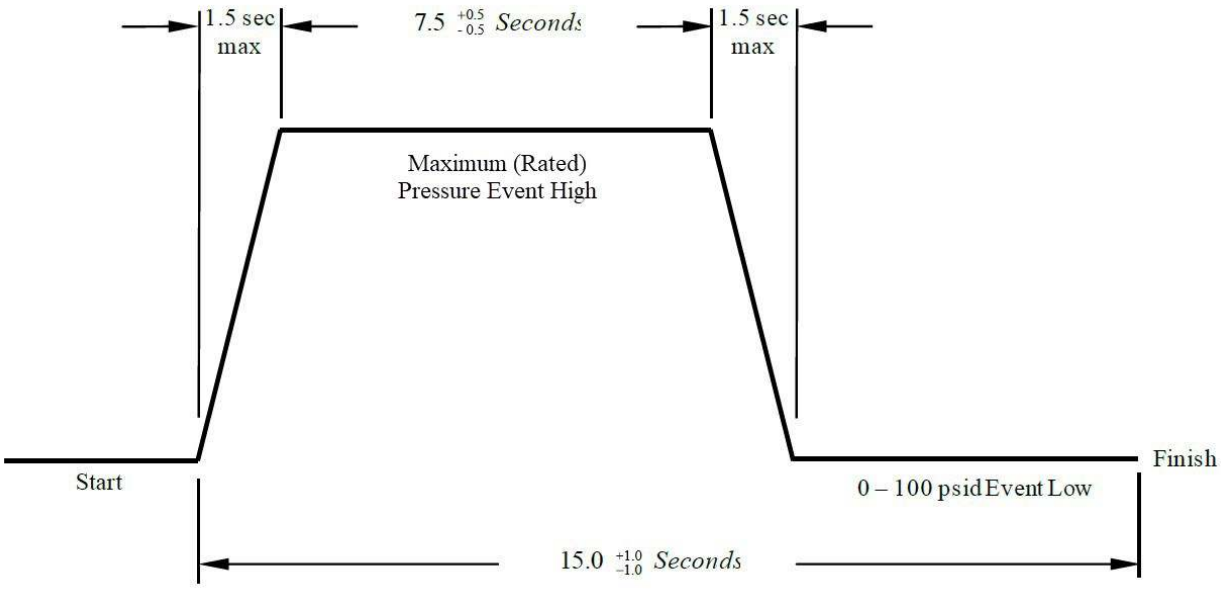


FIGURE 5. Cold start cycle wave shape.

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4.5.2.7 Degree of filtration (filter element efficiency). The degree of filtration for elements shall be determined by the test specified herein. It is of primary importance that the hydraulic fluid and air used in the degree of filtration test be clean and filtered prior to test. Figure 6 shows a degree of filtration test setup with a clean-up device attached.

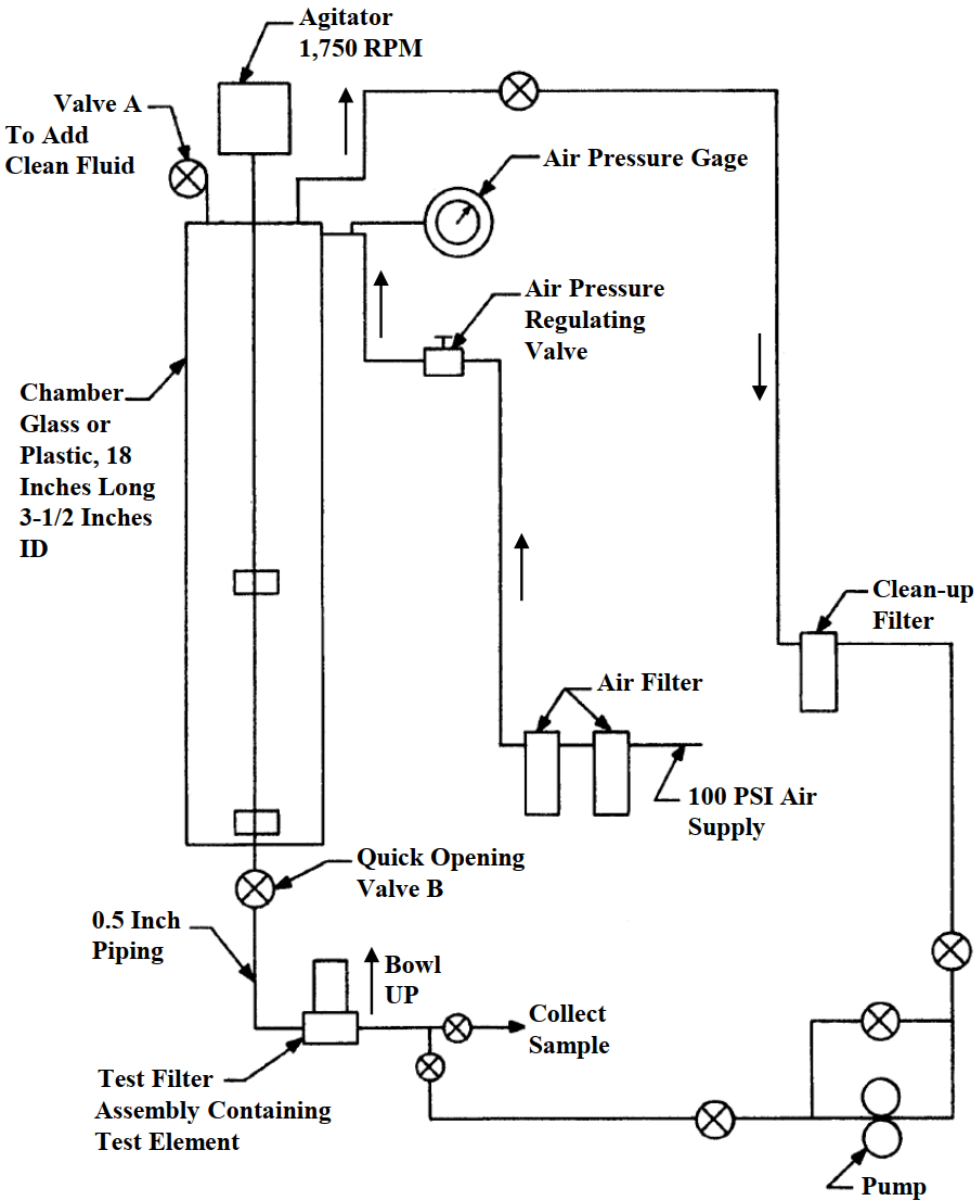


FIGURE 6. Typical apparatus for determining filtration efficiency of line-type element.

4.5.2.7.1 System add and blank values. The test specified in 4.5.2.7.2, with the filter element removed and a free-flow dummy element installed, shall be repeated four times for four

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separate contaminant-add values (A in the 4.5.2.7.2 formula). Hydraulic fluid cleaned to a level that conforms to 4.2.1.1 shall be used for each test. The contaminant collected shall be the add value and shall be used in the calculation of degree of filtration. It shall be the average of these four runs. None of the four add values shall be less than 95 percent by weight of contaminant introduced. To ensure cleanliness of the system and the filter assembly, a system blank value shall be obtained by repeating 4.5.2.7.2 (a) thru (c) with the filter element installed and no contaminant added. This blank value shall be the value C in the following efficiency formula and shall be less than 0.0007 gram.

4.5.2.7.2 Test procedure.

a. A setup shall be made as shown on figure 6 without installing the filter element in the test housing.

b. Flush 2,000 mL of pre-filtered hydraulic fluid through the contaminant mixing chamber and the filter housings, and discard. This operation shall be repeated.

c. The hydraulic fluid of the second flush in (b) shall be checked in accordance with 4.2.1.1.

d. Add 2,000 mL of previously filtered hydraulic fluid through plug valve A. The element shall be installed in the filter housing.

e. Valve B shall be closed.

f. A 5 mL slurry containing contaminant, in accordance with table I of the applicable specification sheet, shall be added to the hydraulic fluid through a small funnel inserted in the plug valve A. The contaminant shall be APM F-9 beads, or A.C. dust and APM F-9 beads as specified.

g. The contaminant shall be distributed uniformly by churning the hydraulic fluid with an agitator for 3 minutes minimum.

h. Plug valve A shall be closed and the glass chamber containing the hydraulic fluid and contaminant shall be pressurized using the air regulator. The air regulator is used to maintain rated flow.

i. Valve B shall be opened and air pressure shall force the hydraulic fluid containing the contaminant through the sample filter assembly. This filtrate shall be collected in a clean 4,000 mL beaker. Using a suitable wash bottle, 750 mL of petroleum ether (boiling point (b.p.) 90 °F to 130 °F) shall then be washed through the contaminant mixing chamber and test filter assembly. The wash fluid shall be collected in the same 4,000 mL beaker. Applicable solvent conforming to MIL-PRF-680 can be used as an alternate to petroleum ether or petroleum naphtha.

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j. A membrane filter 47 mm, diameter disk, absolute 0.8 micron type AA Millipore, or equivalent, shall be washed with 200 mL of pre-filtered petroleum ether and dried to constant weight at 125 °F. The weight shall be accurate to 0.1 milligram (mg). The disk shall be heated at 125 °F for 30 minutes and then cooled 30 minutes in a desiccator.

k. The weighed filter membrane shall be assembled into the filter holder assembly (see figure 5) and fitted to a 4,000 mL vacuum flask connected to a suitable vacuum.

l. All hydraulic fluid passed through the test filter is filtered through the membrane disk. The 4,000 mL beaker shall be washed with 500 mL of pre-filtered naphtha and 500 mL of pre-filtered petroleum ether (b.p. 95 °F to 130 °F) and fluid from the washer beaker shall be passed through the membrane. The filter funnel shall also be washed down with 100 mL of pre-filtered naphtha and 100 mL of pre-filtered petroleum ether.

m. Maintaining vacuum, the top half of the filter funnel assembly shall be removed. The membrane filter shall now be exposed for further washing.

n. With a wash bottle of pre-filtered petroleum ether (b.p. 95 °F to 130 °F), the rim of the membrane filter shall be gently washed to remove traces of hydraulic fluid, being careful not to disturb the cake. Vacuum shall be maintained during this operation.

o. The vacuum shall be shut off and membrane filter removed. It shall be dried to a constant weight at 125 °F and weighed to 0.1 of mg.

The difference between the membrane filter weights before and after the above procedure is the weight of contaminant passing through the test filter, B in the following formula:

$$\text{Filter element efficiency} = \frac{A - (B - C)}{A} \times 100$$

Where:

A = (add value)	= Amount of test contaminant passed through system when there is no filter element in the filter housing.
B = (contaminant value)	= Amount of test contaminant passed through the test filter assembly (element installed).
C = (blank value)	= Amount of contaminant attributed to the test system and test filter assembly when no test contaminant has been added.

The filter element efficiency shall be as specified on the applicable specification sheet.

4.5.2.8 Maximum particle passed (filter element test sequence no. 1). The maximum particle passed shall be determined by the following method:

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a. A degree of filtration (filter element efficiency) test shall be conducted in accordance with the procedure specified in (a), (b), (c), (d), and (e) of 4.5.2.7.2.

b. A 5 mL slurry containing contaminant, in accordance with table I of the applicable specification sheet, shall be added to the hydraulic fluid through a small funnel inserted in the plug valve A.

c. Continue with degree of filtration test in accordance with procedure specified in (g), (h), and (i) of 4.5.2.7.2.

d. All hydraulic fluid passed through the test filter element shall be filtered through a 0.8 micron membrane filter disk AA Millipore, or equivalent (see figure 7). The 4,000 mL beaker shall be washed wit 500 mL of pre-filtered naphtha and 500 mL of pre-filtered petroleum ether (b.p. 95 °F to 130 °F) and hydraulic fluid from the washed beaker shall be passed through the membrane filter disk. The filter funnel shall also be washed down with 100 mL of pre-filtered naphtha and 100 mL of pre-filtered petroleum ether.

e. The entire disk shall be scanned. The largest bead or Carbonyl Iron E particle, as applicable, shall be no greater than the value specified on the applicable specification sheet. This membrane shall be retained for possible scanning in accordance with 4.5.2.8.3.4.

NOTE: In order to obtain an absolute test result, the test set-up downstream of the filter element and all glassware should not have been previously used with glass beads or Carbonyl Iron E.

4.5.2.9 Flow fatigue (filter element test sequence no. 1). Filter elements shall be installed in non-bypass type filter housings and shall be subjected to pressure-flow cycles with hydraulic fluid at minimum of 30 percent of rated flow and at 275 ºF ±5 ºF. A cycle shall consist of increasing the differential pressure across the filter element from zero to the maximum specified and back to zero by first increasing, then decreasing, the flow through the test filter element, which has been loaded with A.C. fine test dust. Peak pressure shall be reached within 25 percent of the cycle (100 to 150 msec) and held for at least 50 percent of each cycle. The test cycle wave is shown on figure 8. The cycling rate, the number of cycles at each differential pressure, and the maximum rated pressure shall be in accordance with the applicable specification sheet. There shall be no evidence of damage as indicated by the bubble point test as a result of this test.

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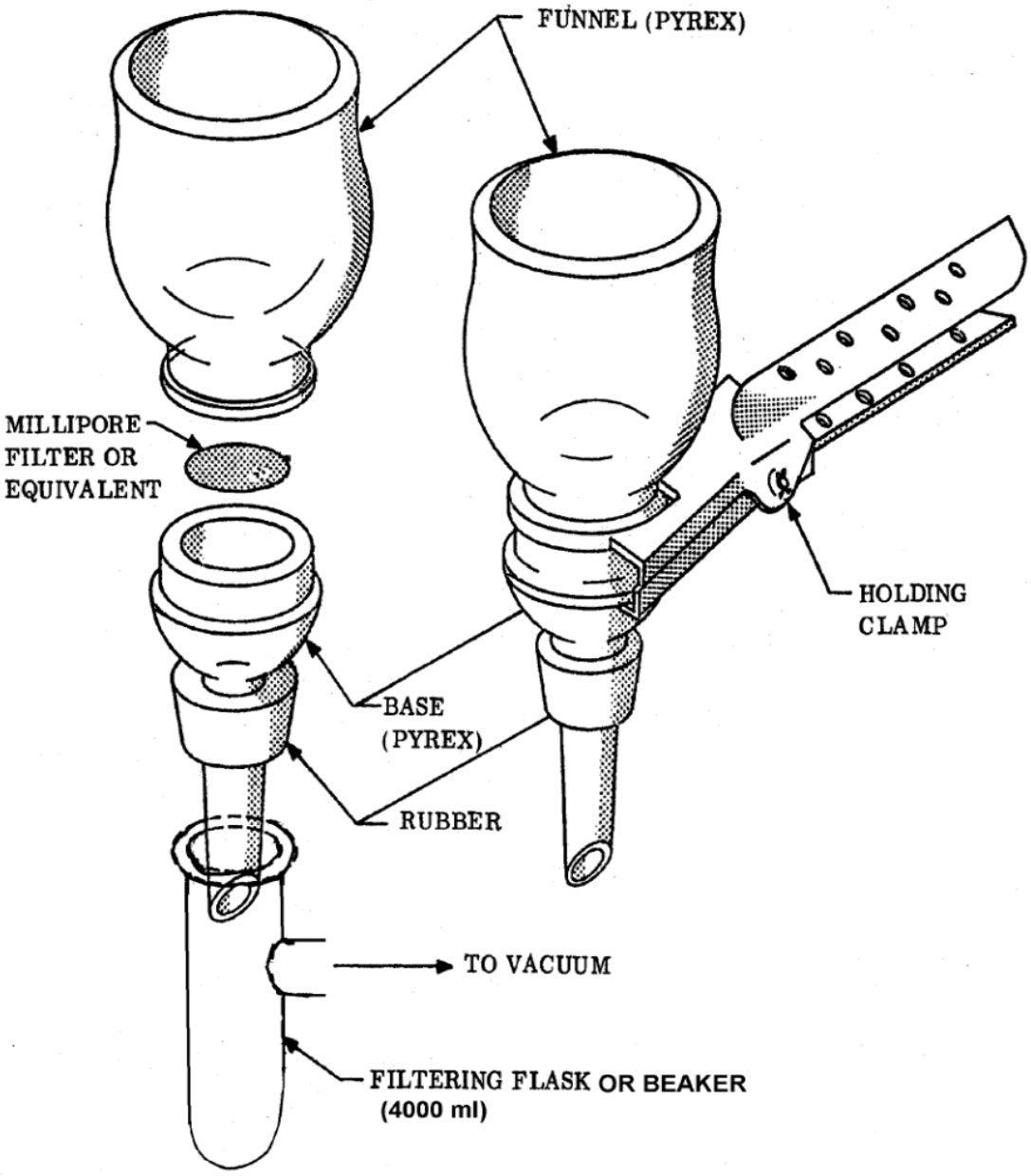


FIGURE 7. Diagram of effluent filtration setup.

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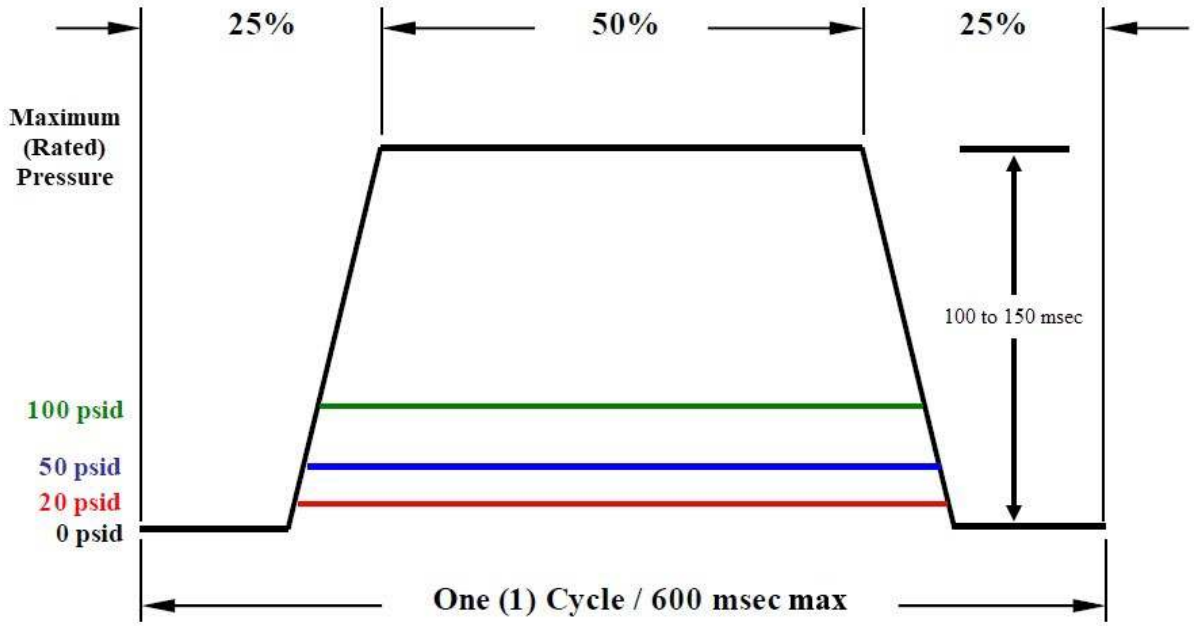


FIGURE 8. Cyclic frequency for flow fatigue test.

4.5.2.10 Pressure buildup and collapse (filter element test sequence no. 2).

4.5.2.10.1 Pressure buildup and collapse for all elements (non-cleanable).

a. The hydraulic test fluid used in the system shown on figure 3 shall be pre-cleaned and checked for cleanliness in accordance with 4.2.1.1. The fluid viscosity shall be a minimum of 13 centistokes for this test.

b. The filter element under test shall be installed in the housing.

c. The pump shall be started, valves A and B opened, and rated flow attained. The cleanup filter shall be by-passed during the entire test. The pressure drop across the filter element shall not exceed the value specified in the applicable specification sheet.

d. Standardized fine air cleaner (A.C.) test dust, in slurry shall be added through the dust valve H (in line with test filter), in the contaminant-add increments applicable to the size of filter element, at 4-minute intervals as specified in table I of the applicable specification sheet. Ten seconds prior to the add interval, begin sampling 100 mL fluid downstream of the test housing. Sampling shall continue as shown on figure 9.

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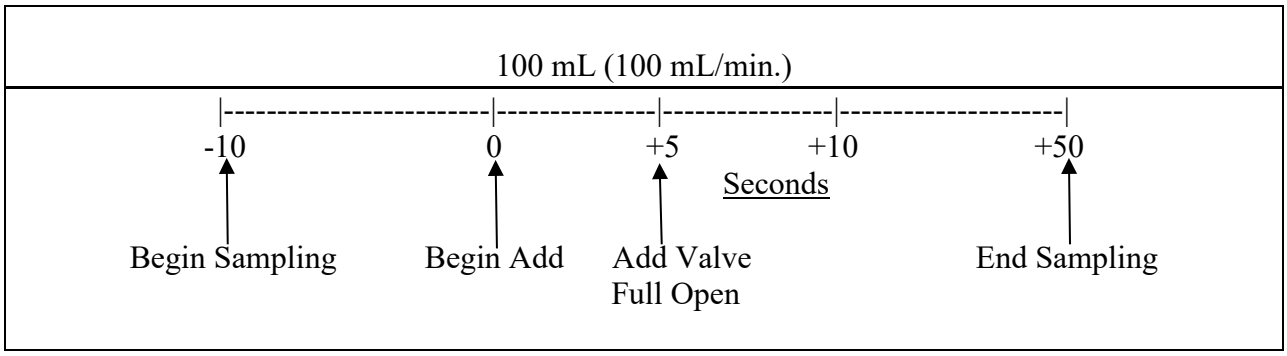


FIGURE 9. Pressure buildup and collapse sampling time base.

Pass the 100 mL sample through a 0.45 micron Millipore membrane filter or equivalent or an automatic particle counter. If a membrane filter is used wash with petroleum ether, dry, identify, and store in a petri dish. Sample fluid with clean element and at every add interval check the differential pressure until a differential pressure of 150 psid is obtained. Count all particles in the following size ranges, in microns: 5-15, 15-25, 25-50, 50-100, and greater than 100. Tabulate to show the count in each size range at each pressure differential. Identify the tabulation with the element part number and serial number. The particle counts shall not exceed SAE-AS4059, class 4. Two minutes after each test dust addition, pressure differential flow and temperature shall be recorded. The differential pressure across the filter element shall not exceed the value specified in table I of the applicable specification sheet when the minimum total A.C. dust shown in table I is added. The pump shall not be stopped during the test. Additional dust shall be added until the minimum collapse pressure specified in the applicable specification sheet (element collapse), is obtained across the element at rated flow. The minimum collapse pressure shall be imposed for two successive times and held for 2 minutes each time. There shall be no sign of structural failure following this test as indicated by a visual examination and bubble point test (see 4.5.2.2.1). For elements loaded to 1.5 times rated pressure (psid), the bubble point test shall be conducted after loading the element to rated pressure (psid).

4.5.2.10.2 Pressure buildup and collapse for 15 micron elements.

4.5.2.10.2.1 Filter element pressure drop (cleanable elements only).

4.5.2.10.2.1.1 Method of test. To ensure filter element cleanability, the filter element shall be subjected to a pressure drop test in a fixture similar to that being used by Naval fleet activities in order to establish a reference element pressure drop. A flow test setup shall be constructed in general accordance with the flow test schematic shown in Drawing 63A109. The test fixture, probe, and plumbing schematic shall be in accordance with the subassembly Drawing 63A109-D-78, 63A109-C-139, and 63A109-C-91. A flow test shall be performed in accordance with 63A109-C-91. The flow test shall be performed using solvent conforming to MIL-PRF-680 or cleaner conforming to MIL-PRF-32295 as a test fluid at the following flow rates:

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Filter element size (M8815/3)	Flow (percent of full scale)
-8	20
-10	30
-12	50
-16	50

4.5.2.10.2.1.2 New element pressure drop value. The value obtained in 4.5.2.10.2.1.1 shall be used to establish the new filter element pressure drop for conformance testing.

4.5.2.10.2.2 Test method.

a. The hydraulic test fluid used in the system shown on figure 3 shall be pre-cleaned as specified in 4.2.1.1. The fluid viscosity shall be a minimum of 13 centistokes for this test. The fluid shall be checked for cleanliness in accordance with 4.2.1.1. The cleanup filter specified in 4.2.1.2 shall be bypassed when the test is begun and will remain bypassed until completion of the tests.

b. The filter element under test shall be installed in the housing.

c. The pump shall be started, valves A and B opened, and rated flow attained. The pressure drop across the filter element shall not exceed the value specified in the applicable specification sheet.

d. Standardized fine air cleaner (A.C.) test dust, in slurry shall be added through the dust valve H (in line with test filter), in the contaminant-add increments applicable to the size of filter element, at 4-minute intervals as specified in table I of the applicable specification sheet. Two minutes after each test dust addition, pressure differential flow and temperature shall be recorded. The differential pressure across the filter element shall not exceed the value specified in table I of the applicable specification sheet when the minimum total A.C. dust shown in table I is added. The pump shall not be stopped during the test. For MIL-F-8815/3 elements, additional dust shall be added until the differential pressure of the rated pressure is obtained across the element at rated flow. The dirt capacity to rated pressure shall be noted and recorded. Rated flow shall be obtained at rated pressure.

e. Cleanable MIL-F-8815/3 elements shall be loaded 10 times to rated pressure and cleaned 10 times. The filter element shall be subjected to the pressure drop test specified in 4.5.2.10.2.1. After final cleaning, the filter shall be subjected to and shall meet all requirements specified in 4.5.2.10.2.2 (a), (b), (c) and (d). Elements shall be cleaned in accordance with the procedure specified in 4.5.2.10.2.3.

f. Circulate fluid at 20 percent of rated flow (minimum), add additional quantities of A.C. dust to the system until the minimum collapse pressure specified in the applicable specification sheet is obtained. The minimum collapse pressure shall be imposed for two successive times and held for 2 minutes each time.

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4.5.2.10.2.2.1 Maximum pressure drop after cleaning value. The value obtained in 4.5.2.10.2.2 (e) shall be indicated on each cleanable filter element in accordance with 3.8.2 (the value shall be subject to the approval of the activity responsible for qualification).

4.5.2.10.2.3 Cleaning procedures. To ensure cleanability and material compatibility with the cleaning solvents, two elements shall be subjected to the tests specified herein. One filter element shall be cleaned in accordance with cleaning procedure I and the second element in accordance with cleaning procedure II.

4.5.2.10.2.3.1 Cleaning procedure I. The filter element shall be cleaned in an ultrasonic cleaner having sufficient power to clean a qualified filter element using the following procedure: (A typical cleaner is one conforming to Drawing 63A109.)

a. Mount the filter element on a rack.

b. Degrease in MIL-PRF-680 solvent or MIL-PRF-32295 cleaner.

c. Ultrasonically clean the filter element in a solution of Turco Surgex, or equivalent, for a period of 30 minutes. The filter element shall be rotated 90 degrees every 5 minutes to facilitate complete cleaning.

d. Rinse the filter element in MIL-PRF-680 solvent or MIL-PRF-32295 cleaner.

e. Ultrasonically clean the filter element in MIL-PRF-680 solvent or MIL-PRF-32295 cleaner for a period of 30 minutes. The MIL-PRF-680 solvent or MIL-PRF-32295 cleaner shall be filtered through a filter element having the filtration characteristics conforming to MIL-PRF-81836. The filter element shall be rotated 90 degrees every 5 minutes during the cleaning period to facilitate complete cleaning.

4.5.2.10.2.3.2 Cleaning procedure II. Filter elements shall be cleaned in accordance with SAE-AIR787, except that ultrasonic cleaning equipment having equivalent power may be substituted, provided that cleaning solutions and cleaning times are the same.

4.5.2.10.3 Filter element pressure drop (cleanable elements) (conformance). The filter element shall undergo a differential pressure test measuring the differential pressure upstream and downstream of the filter under test using a differential pressure transducer or two pressure transducer gauges as specified in ISO 3968, at 100 °F ±5 °F. The pressure drop across the element shall be as specified in the applicable specification sheet. For example, the pressure drop across the element at 6 gpm flow rate shall not be greater than 10 psid.

4.5.2.11 Filter element unloading (filter element test sequence no. 3). The filter elements shall be installed in a test fixture as depicted on figure 3 and loaded until 70 psid ±10 psid is achieved across the element, or as specified within the applicable specification sheet. Pressure, flow, effluent particulate passing and time shall be recorded in accordance with the applicable specification sheet. This loaded condition is considered the terminal differential pressure and should be reached at the completion of the degree of filtration test. Upon reaching the terminal

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differential pressure, the fluid shall be reduced to a pressure of 0 psi and a flow of 0 gpm. The pressure, flow, effluent particulate passing and time shall be recorded. The test filter shall remain at rest with no flow or pressure for a minimum of 30 seconds and maximum of 60 seconds. Following this rest period, pressure and flow rate shall be increased back to the rated flow within 5 seconds. Pressure and flow shall be maintained for 15 minutes and the lowest (first peak) pressure drop shall be recorded. Measure pressure across the element and effluent particulate count passing/releasing from the element and document results. The filters tested shall not exceed 10 percent pressure drop from the first test point. Pressure drop (psid) vs. time (s) and particle count data for >5 and >15 microns (c) shall be provided for every element tested.

4.5.2.12 Relief valve operation.

4.5.2.12.1 Relief valve operation qualification (filter element test sequence no. 3). Filter elements having integral relief valves, see applicable specification sheet, shall be subjected to the following tests. The filter element media shall be blocked and the filter element installed in a suitable housing. By means of a power-driven pump, pressure shall be applied to the inlet port of the filter housing, beginning with a pressure of 20 psi and increasing in increments of 10 psi or less until cracking pressure of the relief valve in the filter element being tested is reached. At each increment, the pressure shall be maintained constant for 5 minutes. At each pressure increment; time shall be not considered “in” or leakage noted until the beginning of the third minute. The leakage rate during the last 3 minutes of each pressure increment shall be noted. The pressure at which the leakage rate through the relief valve amounts to 8 cc (approximately 160 drops) per minutes shall be considered as cracking pressure. Cracking pressure shall occur at 50 ± 5 psi or as required in accordance with the applicable specification sheet. Leakage up to 80 percent of the cracking pressure shall not exceed 4 cc (approximately 80 drops) per minute. Pressure shall be increased until rated flow through the relief valve is obtained. Rated flow shall occur at a pressure differential between inlet and outlet ports not exceeding 80 psi or as required in accordance with the applicable specification sheet. The pressure shall be reduced in 10 psi decrements as above until leakage through the relief valve does not exceed the rate of 10 drops per minute. The resulting pressure shall be considered the relief valve reseating pressure and shall occur at a pressure of 30 psi or above or as required in accordance with the applicable specification sheet.

4.5.2.12.2 Relief valve test (conformance). The filter element relief valve shall be installed in a test block. By means of a power driven pump, pressure shall be applied to the inlet port of the test block, and increased to 40 psi or as required in accordance with the applicable specification sheet. After 3 minutes maximum, the leakage shall not exceed 4 cc/min. The pressure shall then be raised by 10 percent (i.e., to 44 psi). After 3 minutes maximum, the leakage shall not exceed 8 cc/min. The pressure shall then be raised by an additional 25 percent (i.e. to 55 psi). The leakage measured shall exceed 8 cc/min. Increase the flow to 1.5 gpm. The maximum pressure drop shall not exceed 80 psi or as required in accordance with the applicable specification sheet. Reduce the pressure by 45 percent (i.e. to 30 psi) and the leakage measured shall not exceed 4 cc/min. after 3 minutes maximum.

4.5.2.13 Media and abrasion migration (filter element test sequence no. 4).

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4.5.2.13.1 Preparation of filter assembly for blank analysis. Aluminum union fittings of the proper size confirming to SAE-AS5174, with suitable seals and modified to add lock-wire holes, shall be installed in the inlet and outlet ports of the filter housing and lock-wired in place. The unions shall not be removed during the media migration test.

4.5.2.13.2 Blank analysis of filter housing with dummy element without vibration. The housing with a dummy element, shall be installed in the figure 6 system and flushed with pre-filtered hydraulic test fluid (see 4.2.1.1) for 10 minutes at rated flow. The following test shall be conducted:

a. Plug valves A and B shall be closed and the glass chamber containing 2,000 mL of hydraulic fluid shall be pressurized using the air regulator. The air regulator is used to maintain rated flow.

b. Valve B shall be opened and air pressure shall force the hydraulic fluid through the sample filter assembly. This filtrate shall be collected in a clean 4,000 mL beaker. Using a suitable wash bottle, 750 mL of pre-filtered petroleum ether (boiling point (b.p.) 95 ºF to 130 ºF) or hexane solvent shall then be washed through the glass chamber and test filter assembly. The wash fluid shall be collected in the same 4,000 mL beaker.

c. A membrane filter 47 mm diameter disk, absolute 0.8-micron type AA Millipore or equivalent shall be washed with 200 mL of pre-filtered petroleum ether or hexane solvent and dried to constant weight at 125 ºF. The weight shall be accurate to 0.1 milligram (mg). The disk shall be heated at 125 ºF for 30 minutes and then cooled 30 minutes in a desiccator.

d. The weighed filter membrane shall be assembled into the filter holder assembly (see figure 7) and fitted to a 4,000 mL vacuum flask connected to a suitable vacuum.

e. All hydraulic fluid passed through the test dummy filter is filtered through the membrane disk. The 4,000 mL beaker shall be washed with 500 mL of pre-filtered petroleum ether or hexane solvent and fluid from the washed beaker shall be passed through the membrane. The filter funnel shall also be washed down with 100 mL of pre-filtered naphtha and 100 mL of pre-filtered petroleum ether or hexane solvent.

f. Maintaining vacuum, the top half of the filter funnel assembly shall be removed. The membrane filter shall now be exposed for further washing.

g. With a wash bottle of pre-filtered petroleum ether or hexane solvent, the rim of the membrane filter shall be gently washed to remove traces of hydraulic fluid, being careful not to disturb the cake. Vacuum shall be maintained during this operation.

h. The vacuum shall be shut off and the membrane filter removed. It shall be dried to a constant weight at 125 ºF and weighed to 0.1 mg. The difference between the membrane filter weights before and after the above procedure is the blank value C in the abrasion migration formula (see 4.5.2.13.3.3) and shall not exceed 0.0007 gram.

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4.5.2.13.3 Analysis of the filter assembly.

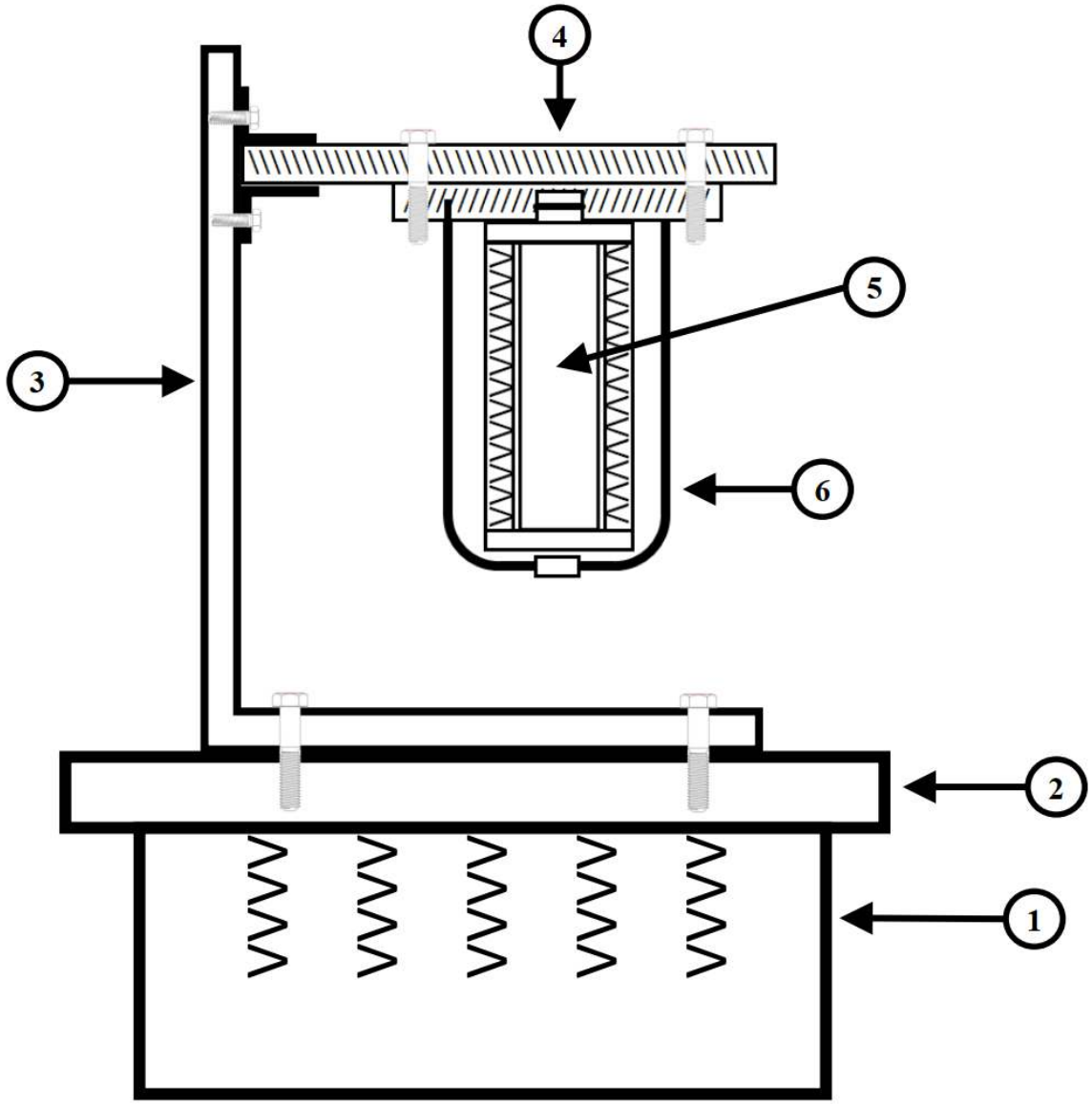
4.5.2.13.3.1 Preparation of filter assembly for vibration analysis. The test element shall be installed in the filter assembly with inlet and outlet union fittings lockwired in place. The filter assembly shall be filled to the top with pre-filtered hydraulic test fluid (see 4.2.1.1); no more than 10 percent of the fluid volume may be drained to allow for expansion. The inlet and outlet union fittings shall be capped with pre-cleaned two-piece aluminum caps (modified to add lockwire holes) conforming to AN929. The test filter assembly shall be mounted in the normal position (bowl at bottom) on a suitable fixture. The bowl shall be completely removed from the head and reinstalled in the head, assuring that only a minimum of fluid be spilled from the bowl. The filter assembly shall be soaked at 275 ºF +0/-10 ºF for 72 hours and then cooled to room temperature (70 ºF to 90 ºF). The cap of the inlet port shall then be removed while the unit is in a horizontal position with the outlet port facing downward. Pre-filtered hydraulic fluid (see 4.2.1.1) shall then be added to completely fill the unit. Port caps and bowl shall then be lockwired to the filter head.

4.5.2.13.3.2 Media and abrasion migration. A sinusoidal vibration test shall be conducted on the filter assembly at room temperature in each of the filter assemblies' three mutually perpendicular axes. One of these axes shall be that which is parallel to the exit flow path of the filter element. The filter assembly shall be mounted on a resonance-free fixture for test (see figure 10). The vibration input accelerometer shall be mounted on the fixture close to the mounting point of the filter assembly to the fixture. A second accelerometer shall be mounted directly on the filter assembly for monitoring resonant frequencies. The rate of change of frequency shall be logarithmic.

Vibration tolerances

Amplitude	±10 percent
Frequency	±2 percent
Acceleration	±10 percent

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1. Vibration Shaker Base
2. Vibration Table
3. Vibration Sturdy Mounting Fixture (Bracket)
4. Filter Mounting Bracket
5. Filter Element Under Test
6. Filter Bowl (Top & Inlets Capped Off)

FIGURE 10. Media migration setup with vibration.

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The following tests shall be conducted:

a. Resonance search - resonant frequencies of the filter assembly shall be determined by varying the frequency of applied vibration slowly through the range specified in table III, part 1, at reduced levels but with sufficient amplitude to excite them. This shall be repeated for each of the three axes.

b. Resonance dwell - the filter assembly shall be vibrated along each axis at the four most severe resonant frequencies (determined in the resonance search) for 30 minutes at each resonance. The vibration levels and test times shall be in table III, part 1.

c. Cycling test - the filter assembly shall be vibrated along each axis in accordance with the test levels, frequency range and times from table III, parts 1 and 2. The frequency of applied vibration shall be swept over the frequency range of 5 to 2,000 to 5 Hertz (Hz) logarithmically. The time to sweep from 5 to 2,000 to 5 Hz shall be approximately 20 minutes. If a change in the resonant frequency occurs during the test, its time of occurrence shall be recorded and immediately the frequency shall be adjusted to maintain the peak resonance condition. The final resonant frequency shall be recorded.

TABLE III. Test procedure and time schedule.

	Vibration Level					
Part 1	0.1 inch double amplitude 1.0 g 0.036 inch double amplitude 10.0 g					
	Frequency Band, Hz					
	5 to 14 14 to 23 23 to 74 74 to 2000					
Part 2	0.2 inch double amplitude 1.0 g 0.06 inch double amplitude 20.0 g					
	5 to 10 10 to 18 18 to 81 81 to 2000					
	Vibration Time Schedule - (Time Per One Axis)					
Part 1	Number of resonances	0	1	2	3	4
	Total Vibration Time at each Resonance	0	0.5 hr	1 hr	1.5 hr	2 hr
	Cycling Time	2 hr	1.5 hr	1 hr	0.5 hr	0
Part 2	30 minutes cycling per axis - no resonance dwell					

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4.5.2.13.3.3 Migration analysis. The caps shall be removed from the port unions and the test specified in 4.5.2.13.2.1 shall be performed on the filter assembly. The value obtained represents the analysis value A in the following migration formula:

$$X = A-C$$

Where A = Analysis value specified in 4.5.2.13.3.3 (in mg)
C = Blank value of housing without vibration specified in 4.5.2.13.2.1 (in mg)
X = Migration analysis value (in mg)

The migration analysis value shall not exceed the values listed in table I of the applicable specification sheet.

4.5.2.13.3.4 Filter element media migration. After completion of the weights analysis, the analysis membrane from 4.5.2.13.3.3 shall be examined for evidence of filter media migration using the procedure specified in SAE-AS598. There shall be no media migration identifiable as coming from the filter element medium. The membrane filter used for this test shall be of contrasting color to the filter element medium. A scanning electron microscope (SEM) with a minimum of 100x magnification shall be used to accurately analyze the material collected on the membrane and to verify that no particulates can be traceable to the filter medium. In the event the abrasion material masks possible filter media migration, the membrane retained from the maximum particle passed test (see 4.5.2.8) shall be scanned for evidence of filter media migration. There shall be no migration identifiable as coming from the filter medium. Photograph(s) from the SEM shall be documented.

4.5.2.14 Pressure drop (filter element test sequence no. 5).

4.5.2.14.1 Visual examination. The filter element shall be visually examined for any visible defects or damage that may have occurred during movement of the element.

4.5.2.14.2 Pressure drop test. Each element shall undergo a second pressure drop test as specified in 4.5.2.5. The pressure differential between the initial pressure drop values and this pressure drop shall not exceed 25 percent and shall not exceed the value specified in the applicable specification sheet.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful but is not mandatory.)

6.1 Intended use. The filter assemblies and filter elements covered by this specification are intended for use in aircraft and missile hydraulic systems covered by SAE-AS5440 and SAE-AS8775 operating with hydraulic fluid conforming to MIL-PRF-5606, MIL-PRF-83282, MIL-PRF-87257 at nominal operating pressures 1,500 psi to 5,000 psi and temperature range -65 ºF to +275 ºF.

6.1.1 Military unique application. The filter assemblies and filter elements covered by this specification are specifically designed and dimensioned uniquely for military aircraft and missile hydraulic systems as specified herein and in the specification sheets and not available for use with commercial aircraft or other non-military equipment systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. PIN - part identification number (see MIL-DTL-8815 Supplement 1).

c. Filter assembly style and pressure rating (see 1.2).

d. Element type, class, micron rating, and size (see 1.2).

e. Qualified Product Database (QPD) and activity responsible for qualification inspection.

f. Packaging and packing (see sections 5.1 and 6.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 8815 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Air Systems Command, Naval Air Warfare Center, Attn: Code BL32600, System Standardization, JBMDL, NJ 08733; however, information pertaining to qualification of products may be obtained from the Naval Air Systems Command, Code AB24200, Building 2187, 48110 Shaw Road, Patuxent River MD 20670. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.4 Excess. Elements manufactured in excess of contract quantity may be placed in stock and accepted for later contracts (within 3 years) based on prior tests of the specific lot.

6.5 A.C. fine dust. This dust is available from Naval Air Systems Command, Code AB24200, Building 2187, 48110 Shaw Road, Patuxent River, MD 20670.

6.6 Subject term (key word) listing.

Cold start
Differential pressure indicator
Hydraulics

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - AS
Air Force - 11
DLA - GS

Preparing activity:
Navy - AS
Project 1650-2022-002

Review activities:
Navy - MC, SA, SH
Air Force - 71
DLA - CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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